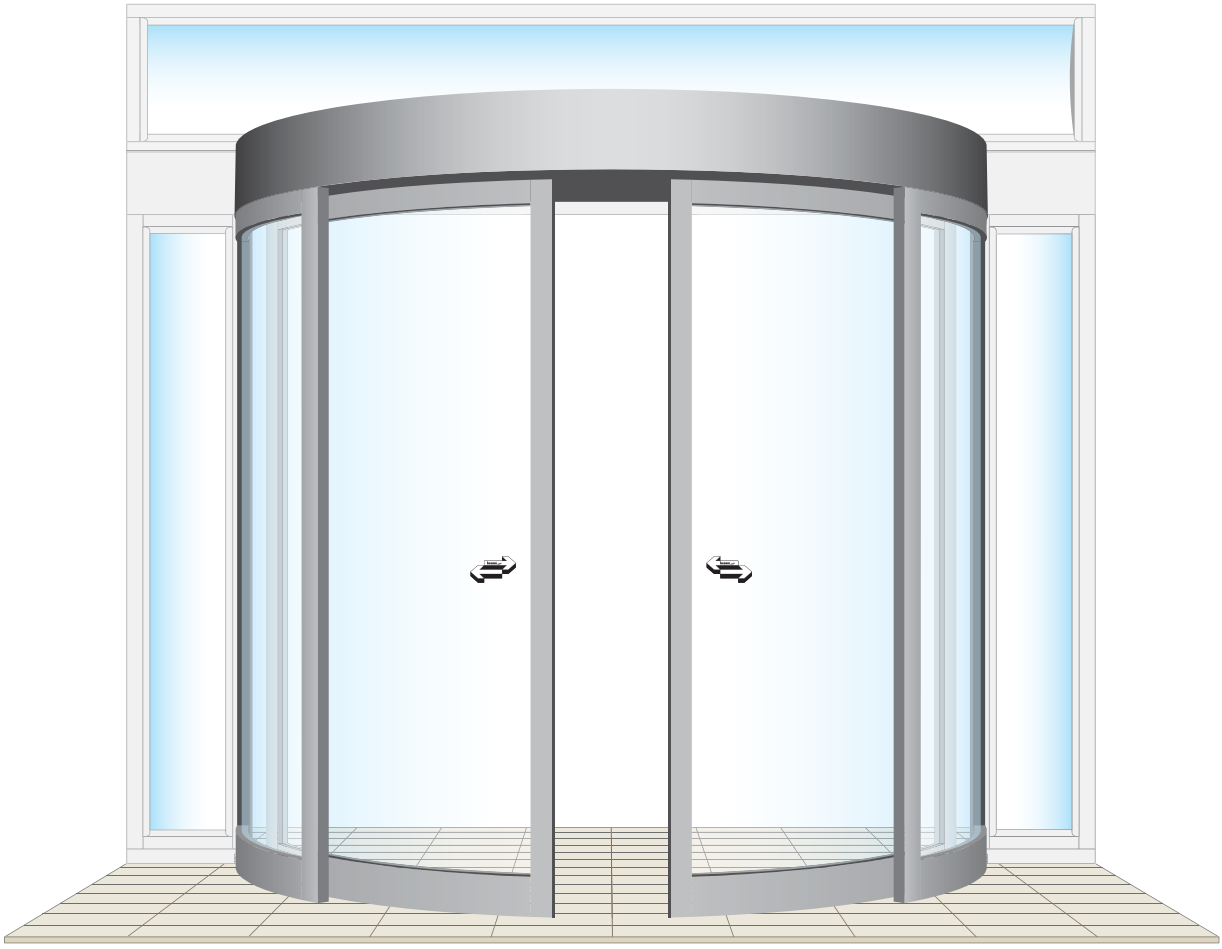
Besam Curved Sliding Door System CMD

Installation and Service Manual



FAA108-EI-2.0 – Issue 2007-06-13

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Table of content

1 Revision . . . 5
2 Important information . . . 6
2.0.1 Important safety instructions for use . . . 6
2.0.2 Important safety instructions for installation . . . 6
2.0.3 Radio and television reception . . . 6
2.0.4 Environment . . . 6
3 Introduction . . . 7
4 Design and function description . . . 8
4.1 Design . . . 8
4.1.1 Main parts . . . 8
4.1.2 Operator . . . 9
4.2 Function . . . 10
4.2.1 Integrated safety detection . . . 10
4.2.2 Safety detection with presence sensor . . . 10
4.2.3 Emergency escape . . . 11
5 Technical specification . . . 12
6 Models . . . 13
7 Mechanical installation . . . 14
7.1 Floor surface . . . 14
7.2 Fixing rails . . . 15
7.3 Outer walls and Support beam . . . 16
7.4 Hanging the door leaves . . . 17
8 Electrical connections . . . 18
8.1 Mains supply and connection . . . 18
8.2 Control unit . . . 19
8.2.1 Contacts for connection of standard units . . . 19
8.2.2 Terminal block for connection of accessories . . . 19
8.2.3 Function selector (FS) used to select special operating functions . . . 20
8.2.4 Potentiometers and learn button . . . 21
8.2.5 Connection of programme selectors . . . 22
8.2.6 Connection of activation units . . . 23
8.3 Extension unit . . . 24
8.3.1 Extension unit, EXU-1 . . . 24
8.3.2 Replacement of the extension unit EXU-1 . . . 25
9 Start-up . . . 26
10 Programme selectors . . . 29
11 Troubleshooting . . . 30
11.1 Mechanical checking and remedies . . . 30
11.2 LED indication and CT Error codes . . . 30
11.2.1 Normal operation/Non-critical errors . . . 30
11.2.2 Power failure (no error code) . . . 30
11.2.3 LED indication . . . 30
11.2.4 Sensor Error . . . 31

11.2.5 Emergency Unit Error 31
11.2.6 CU error 32
11.2.7 Motor / Encoder error 33
11.2.8 Lock error 33
11.2.9 Motor Temperature High 33
11.2.10 on-Critical errors 34
11.3 After remedy or replacement the operator has to be checked as follows: 34
12 Maintenance/Service 35
13 Accessories 38

1 Revision

Page	Revision
22	PS-5M connection, changed from “Auto width” to “Auto partial”
29	Illustration revised, “Auto width” changed to “Auto partial”

2 Important information

2.0.1 Important safety instructions for use

Warning – It is important for the safety of persons to follow these instructions. Save these instructions.

2.0.2 Important safety instructions for installation

Each product is submitted with instructions for how to make the complete installed automatic door safe in use. For further information see STI 04-014.

Warning – Incorrect installation or adjustments can lead to severe injury. Follow and save these instructions.

2.0.3 Radio and television reception

This equipment generates and uses radio frequency energy, and if not installed and used properly, that is in strict accordance with the manufacturer's instructions, may cause interference to radio and television reception. It has been designed to comply with the emission limits in accordance with EN 50081-1 (US market FCC Part 15), which are designed to provide reasonable protection against such interference in a residential installation. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Re-orient the receiving antenna.
- Relocate the receiver with respect to the equipment.
- Move the receiver away from the equipment.
- Plug the receiver into a different outlet so that equipment and receiver are on different branch circuits.
- If necessary, the user should consult the dealer or an experienced radio/television technician for additional suggestions.

2.0.4 Environment

This operator may be equipped with batteries containing materials, which are hazardous to the environment. Remove the batteries from the operator before it is scrapped. The batteries must be disposed of safely.

3 Introduction

This manual contains necessary details and instructions for the installation,maintenance and service of Besam Curved Sliding Door System CMD.

Besam Curved Sliding Door System CMD ensures all-around safety. It can be combined with the full range of Besam safety units, such as presence and motion sensors.

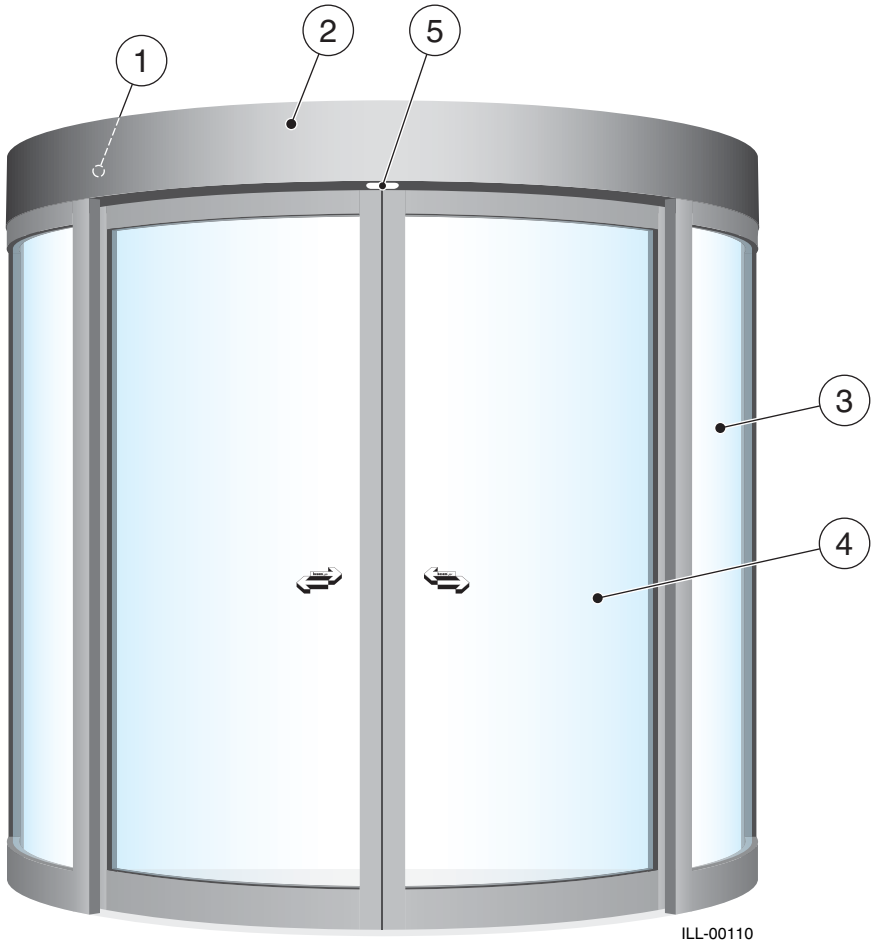
Each installation is however unique and must therefore be equipped and adjusted for the application-relevant safety requirements, just like maintenance must be performed as specified for the selected material.

4 Design and function description

4.1 Design

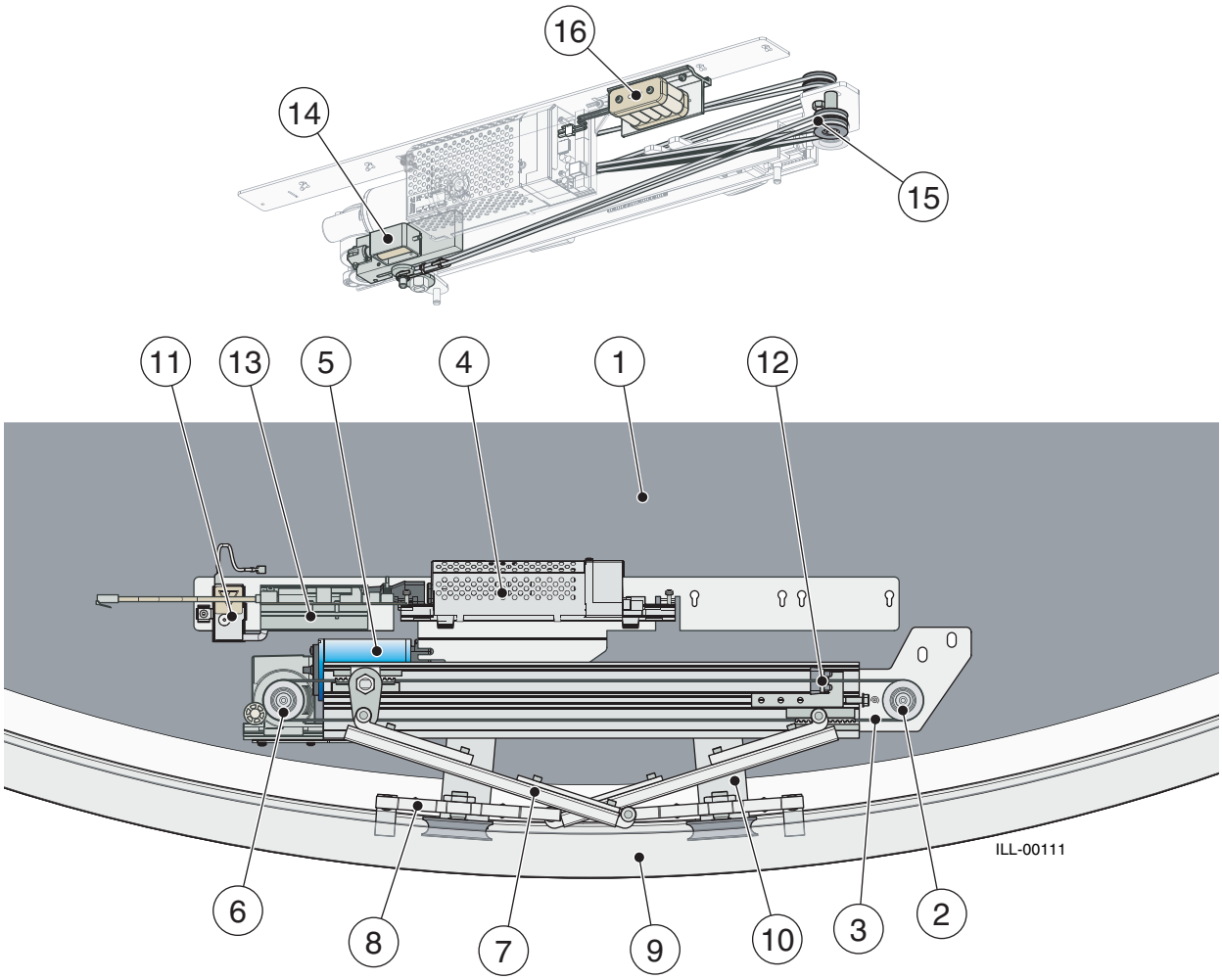
The sliding door operator Besam UniSlide C works electromechanically. The motor, control unit, transmission – and optional emergency unit and electromechanical locking device – are all assembled in a support beam with integrated cover. The motor and gear box transmit movement to the door leaves by means of a tooth belt. The door leaf is fitted to a door adapter/carriage wheel fitting and hangs on a sliding track. The guiding at the bottom is carried out by means of floor guides.

4.1.1 Main parts



No.	Description
1	Operator cover
2	Operator support beam (incl. sliding track)
3	Screens left and right
4	Doors left and right (incl. floor guide)
5	Presence sensor

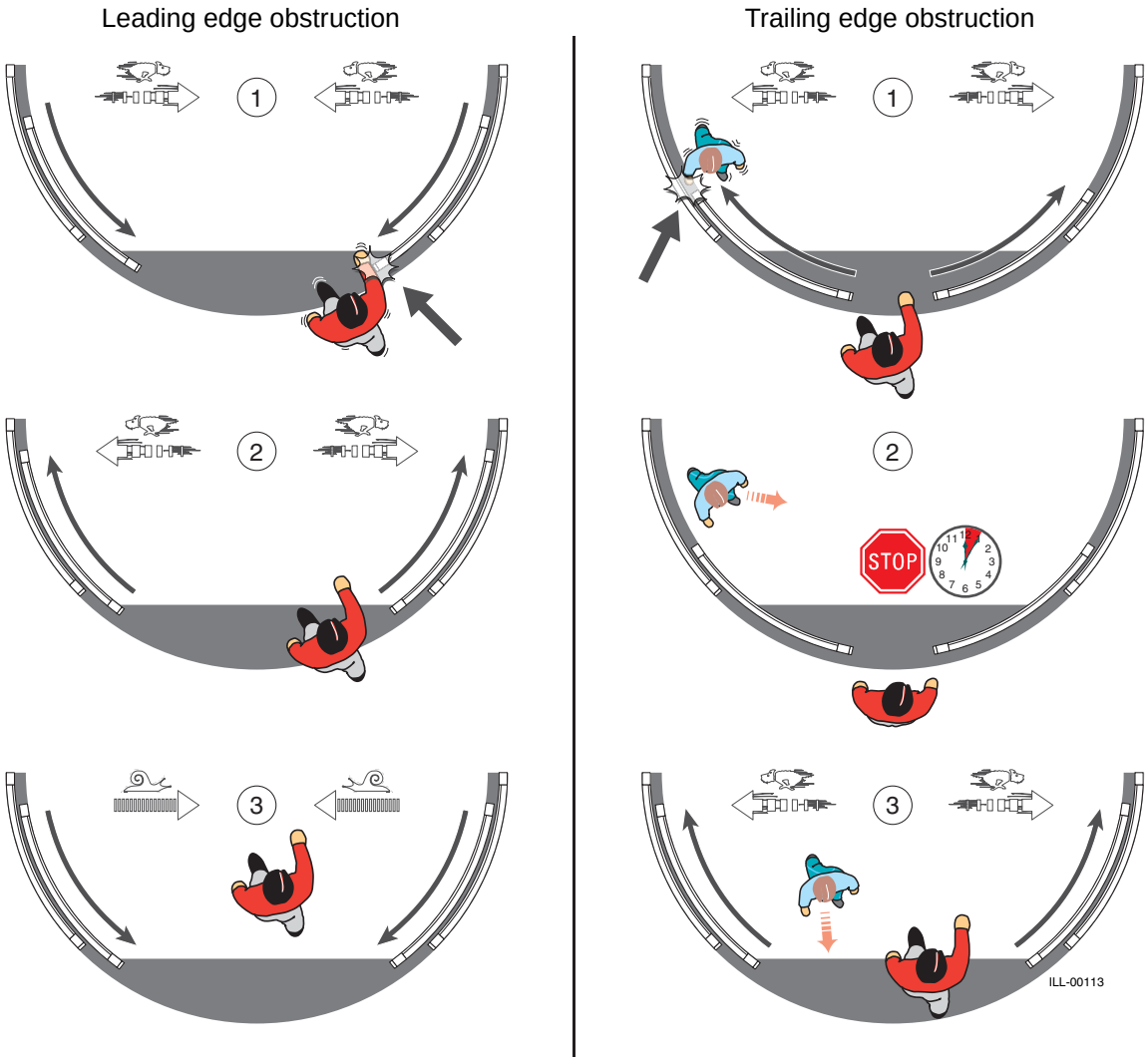
4.1.2 Operator



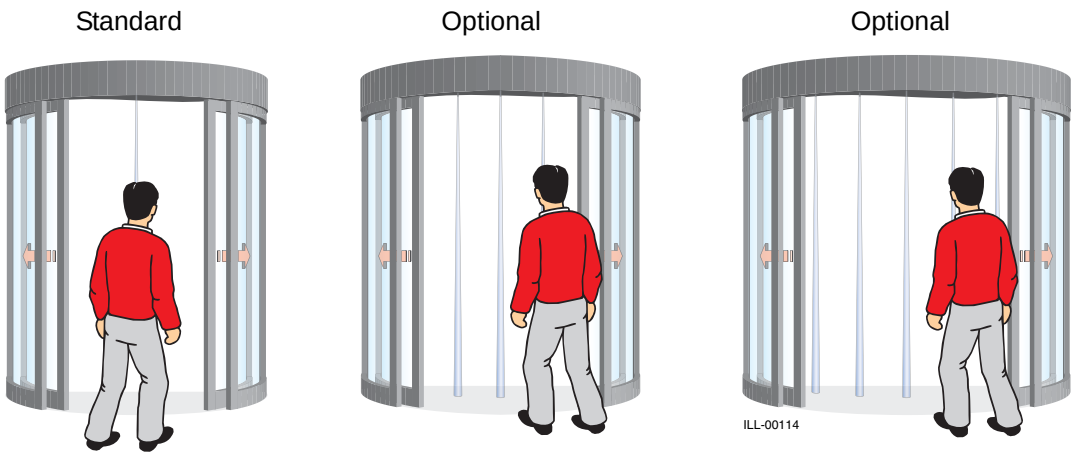
No.	Description
1	Cover
2	Tension wheel
3	Tooth belt
4	Control unit
5	Drive unit
6	Drive wheel
7	Drive arm
8	Carriage wheel fitting
9	Operator support beam (incl. sliding track)
10	Bracket for drive module
11	Mains power connection block
12	Door stop
13	Programme selector board
14	Electromechanicl locking device, option
15	Mechanical Emergency UniT (MEU), option
16	Electrical Emergency Unit (EEU), option

4.2 Function

4.2.1 Integrated safety detection



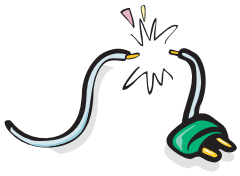
4.2.2 Safety detection with presence sensor



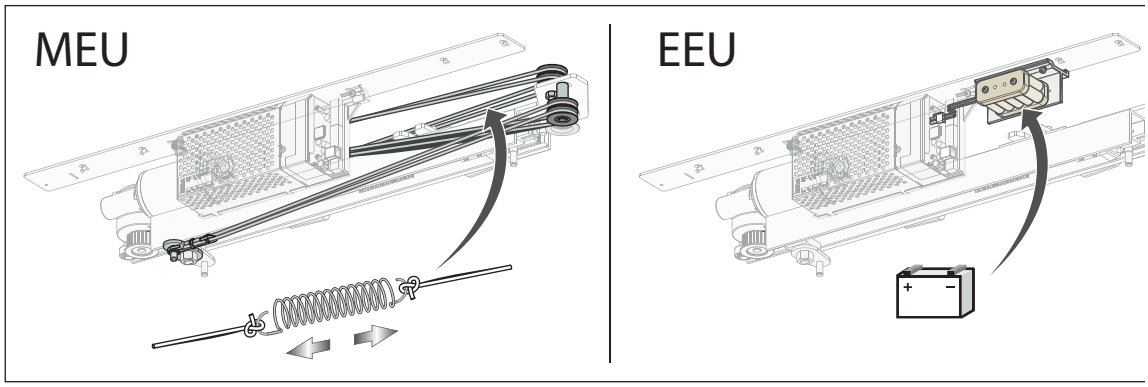
4.2.3 Emergency escape

- The operator can be fitted with different emergency escape units to ensure a safe evacuation of the building: Mechanical emergency unit (MEU) or Electrical emergency unit (EEU). The units can also be interfaced with the fire alarm or smoke detector.

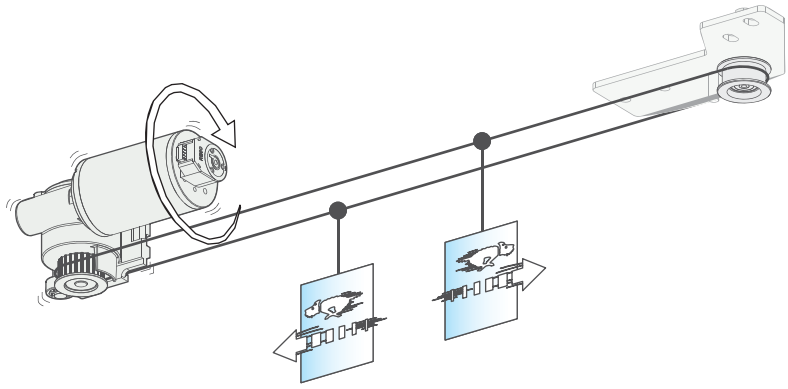
1



2



3



Every 4 h the operator performs a test of the ermegency opening. If the battery (MEU) or the elastic cord (EEU) opens the door within limited time, the test is successful and the door resumes the function set by the programme selector. Otherwise the door remains open.

Note! The test is never performed in programme selector setting “Open”. In setting “Off” it can be selected.

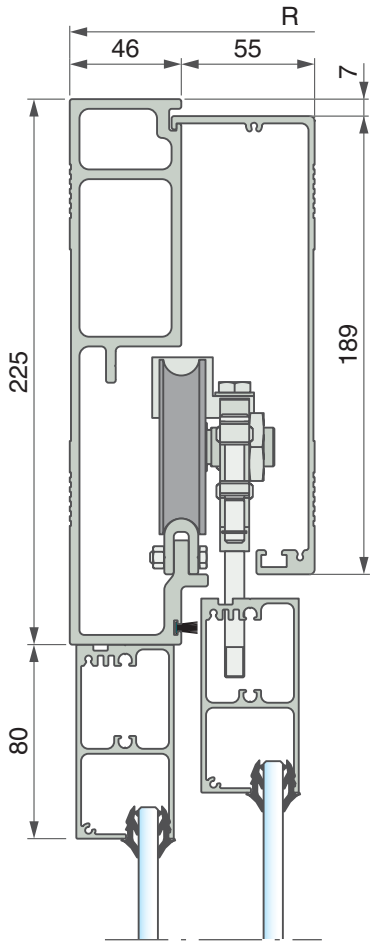
The electrical emergency function can also be used to close the door in the event of power failure.The fire authorities sometimes require this to stop fire or smoke from spreading throughout the building.

5 Technical specification

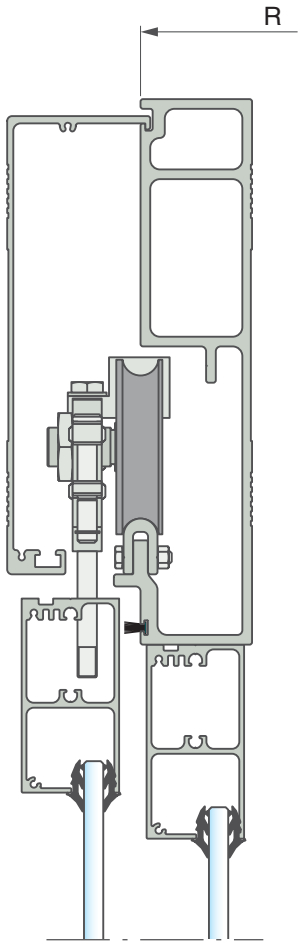
Mains power supply	100 V AC -15% to 240 V AC +10% 50/60 Hz fuse 10 AT **Note!** Switch with clearly marked off-position, having a contact separation of at least 3 mm in all poles, must be incorporated in the fixed wiring, be installed at a minimum height of 1.5 m and not be accessible for the public.
Power consumption	Max. 250 W
Auxiliary voltage	24 V DC, 640 mA
Control unit fuse	6,3 AT
Recommended max. door weight	100 kg/leaf
Minimum radius	900 mm
Opening and closing speed	Variable up to approx. 1.4 m/s
Hold open time	0-60 s
Ambient temperature	-20°C to +50°C
Relative humidity (non-condensing)	Max. 85%
Approvals	Sliding doors in escape routes (SP/TÜV) with the following options: MEU, PSMB-5, LDP, LDB, R3. Monitoring values are factory set and can only be changed by an authorized Besam service centre.
Door operator	For indoor use only

6 Models

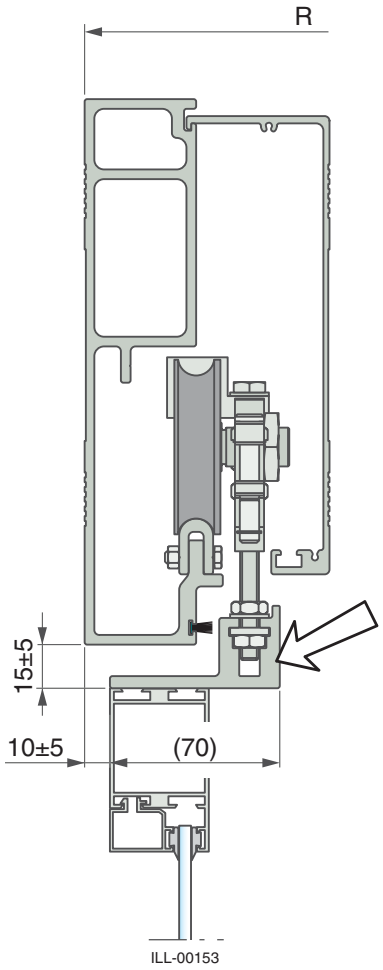
Internal door



External door

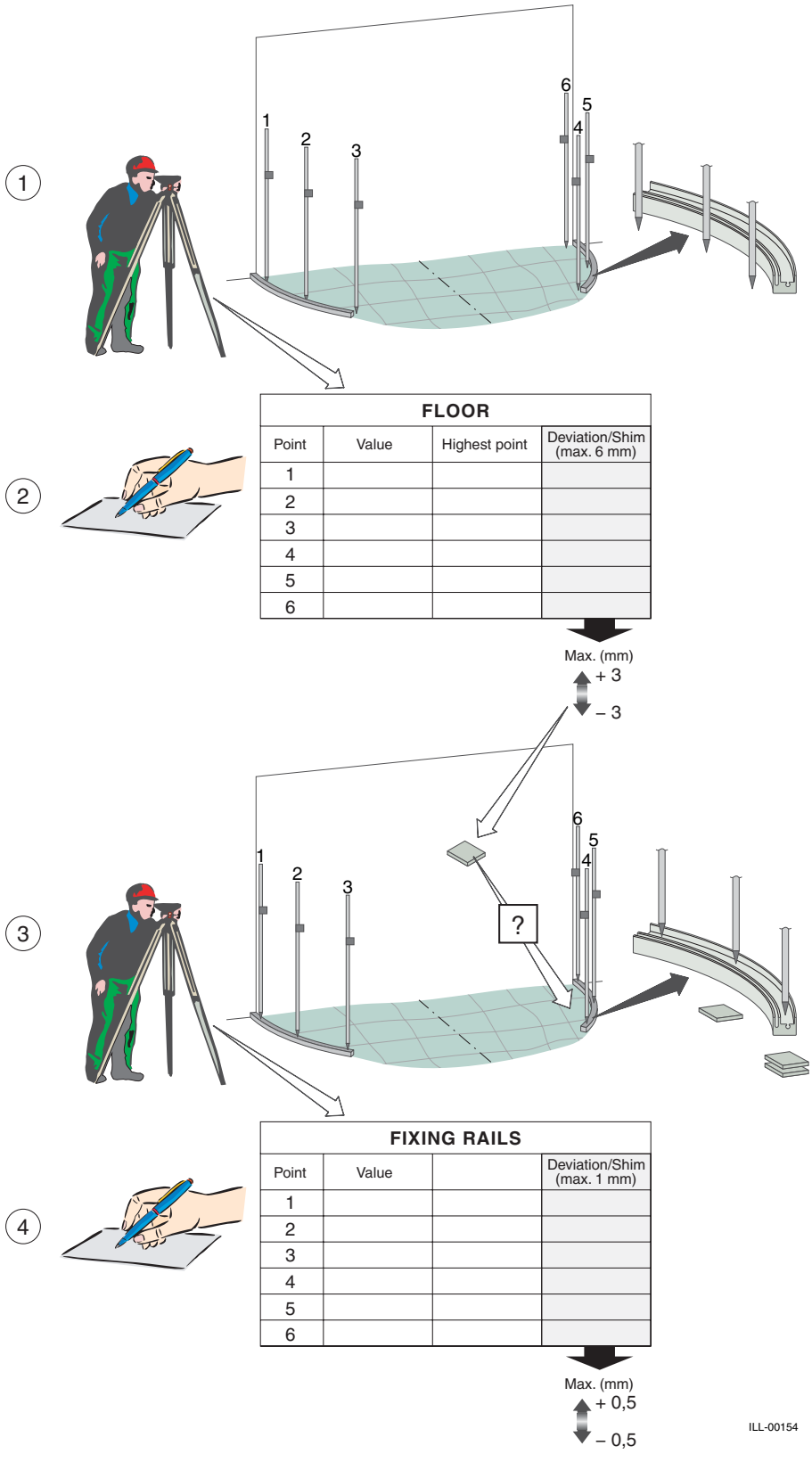


Door adaptor for third party doors

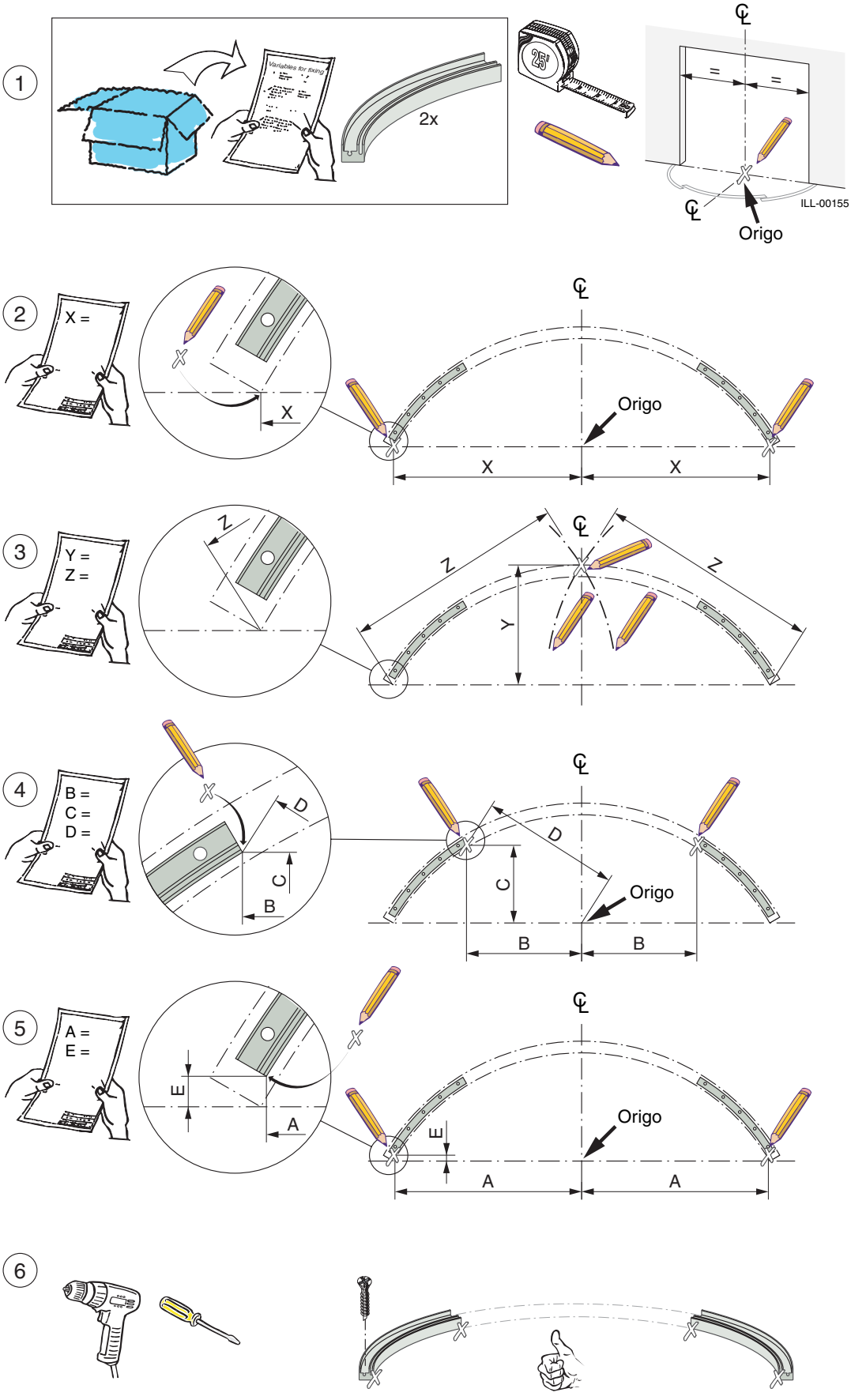


7 Mechanical installation

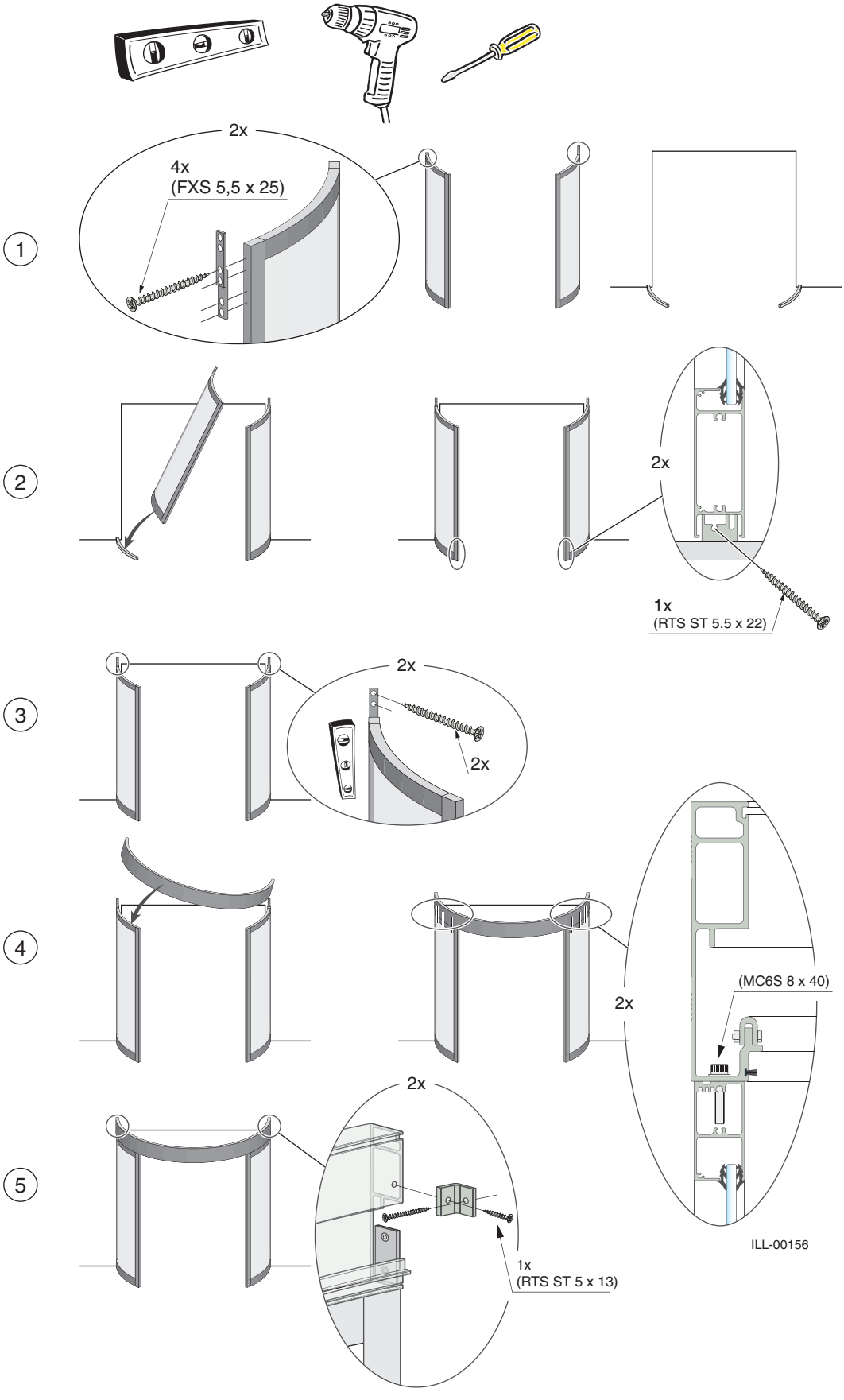
7.1 Floor surface



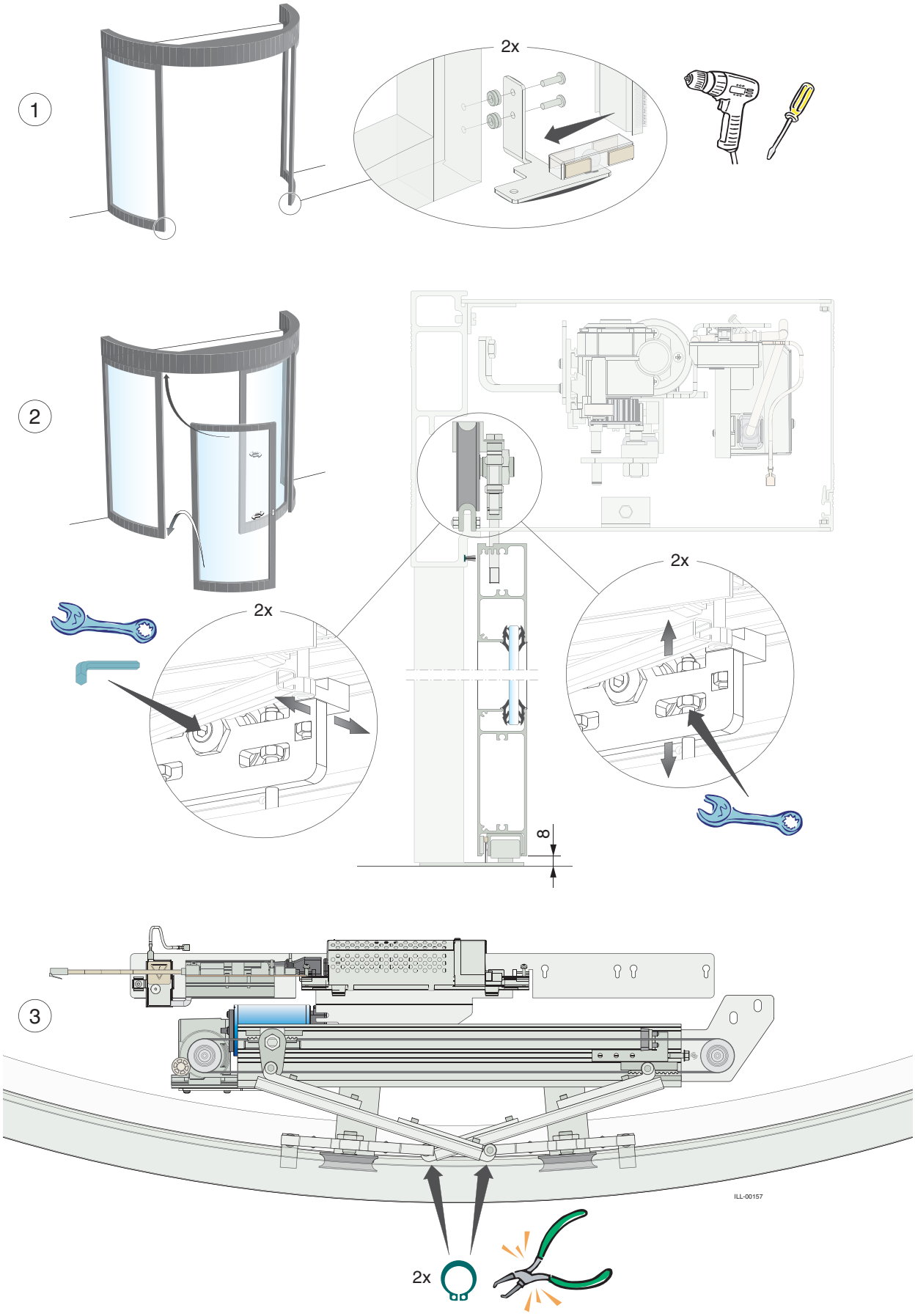
7.2 Fixing rails



7.3 Outer walls and Support beam



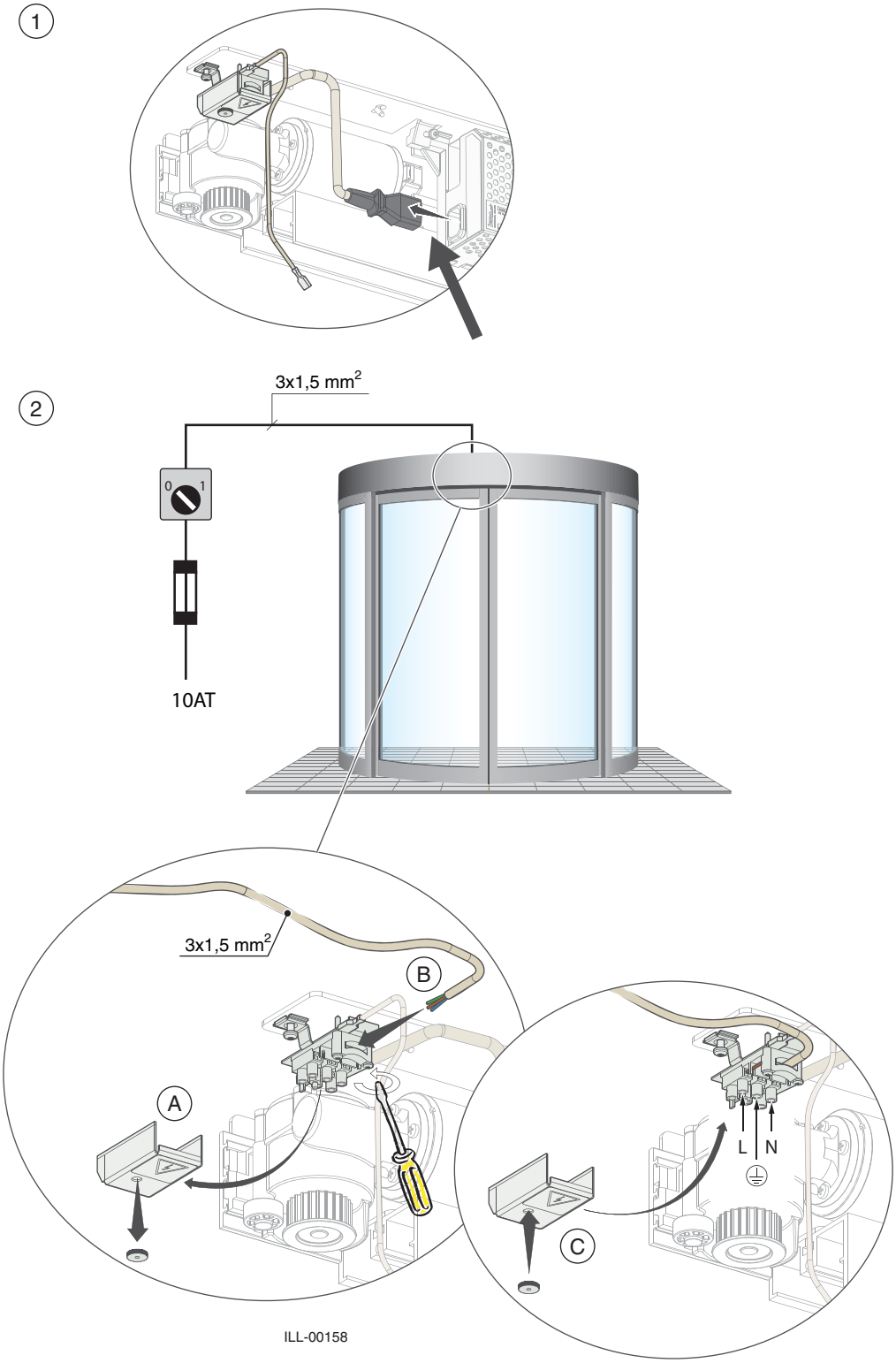
7.4 Hanging the door leaves



8 Electrical connections

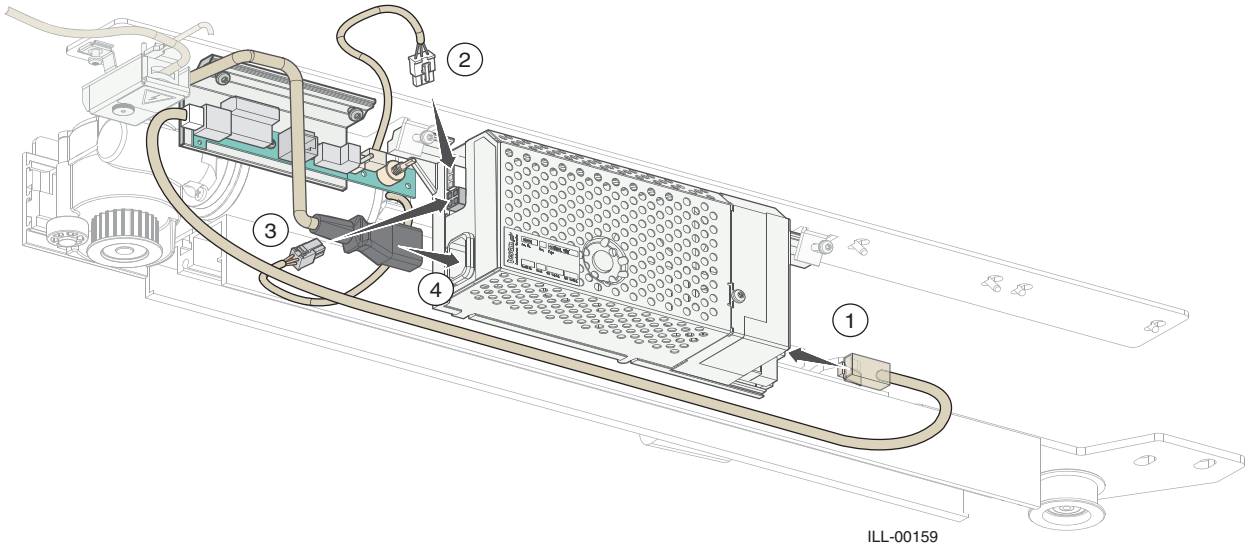
Note! During any work with the electrical connections the
- **mains power** and the
- **electrical emergency unit must be disconnected.**

8.1 Mains supply and connection

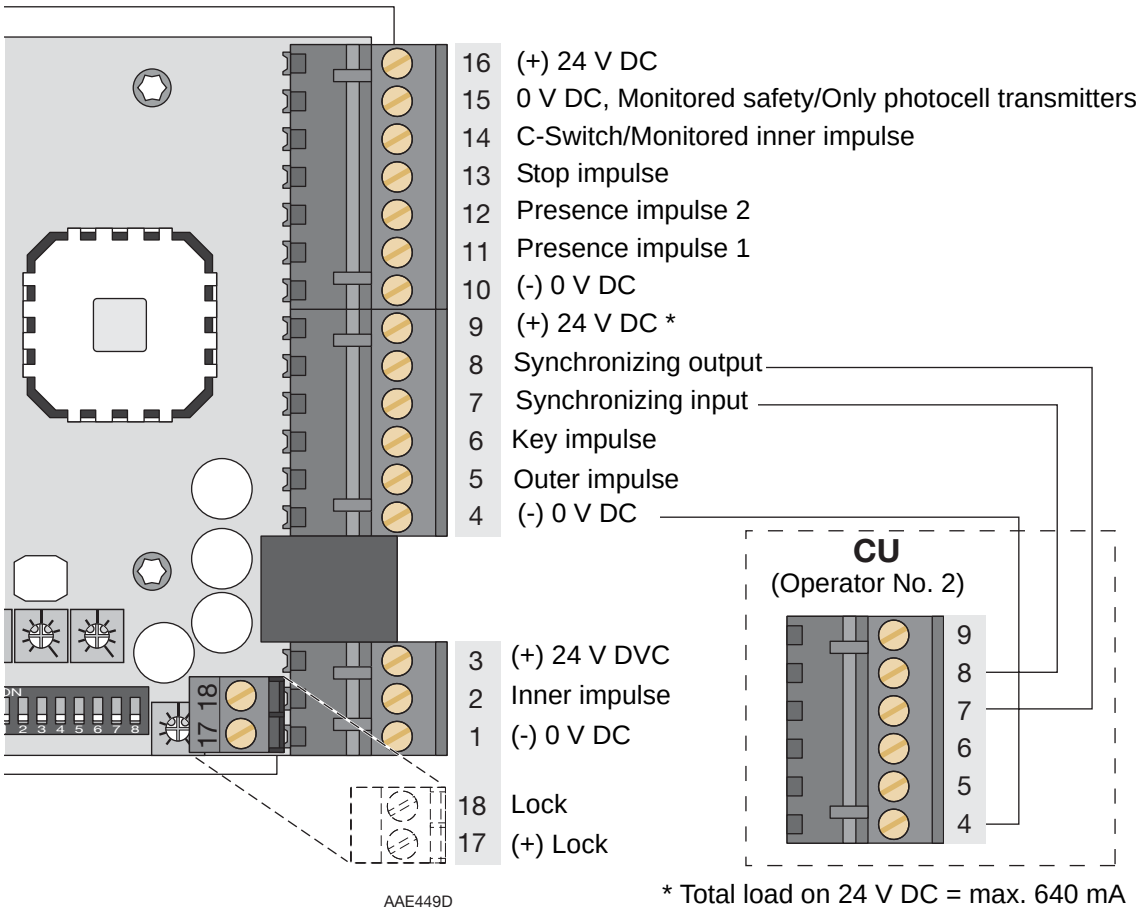


8.2 Control unit

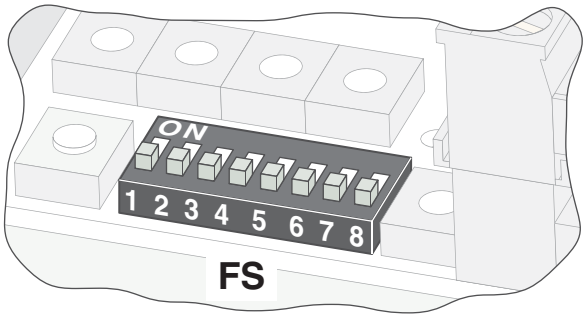
8.2.1 Contacts for connection of standard units



8.2.2 Terminal block for connection of accessories

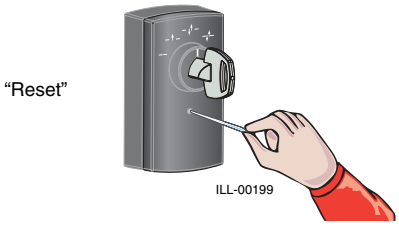


8.2.3 Function selector (FS) used to select special operating functions

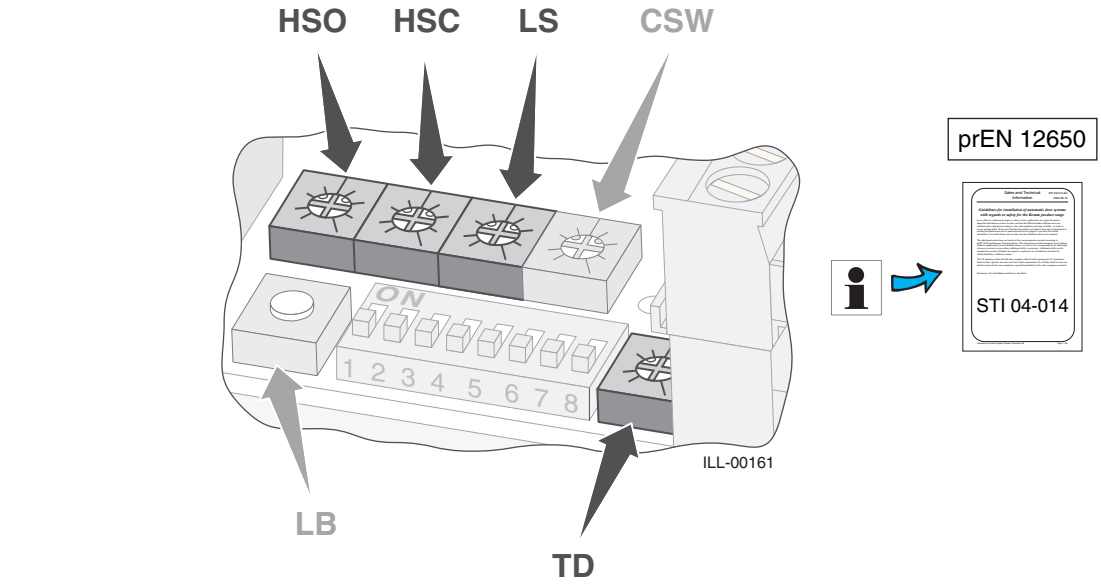


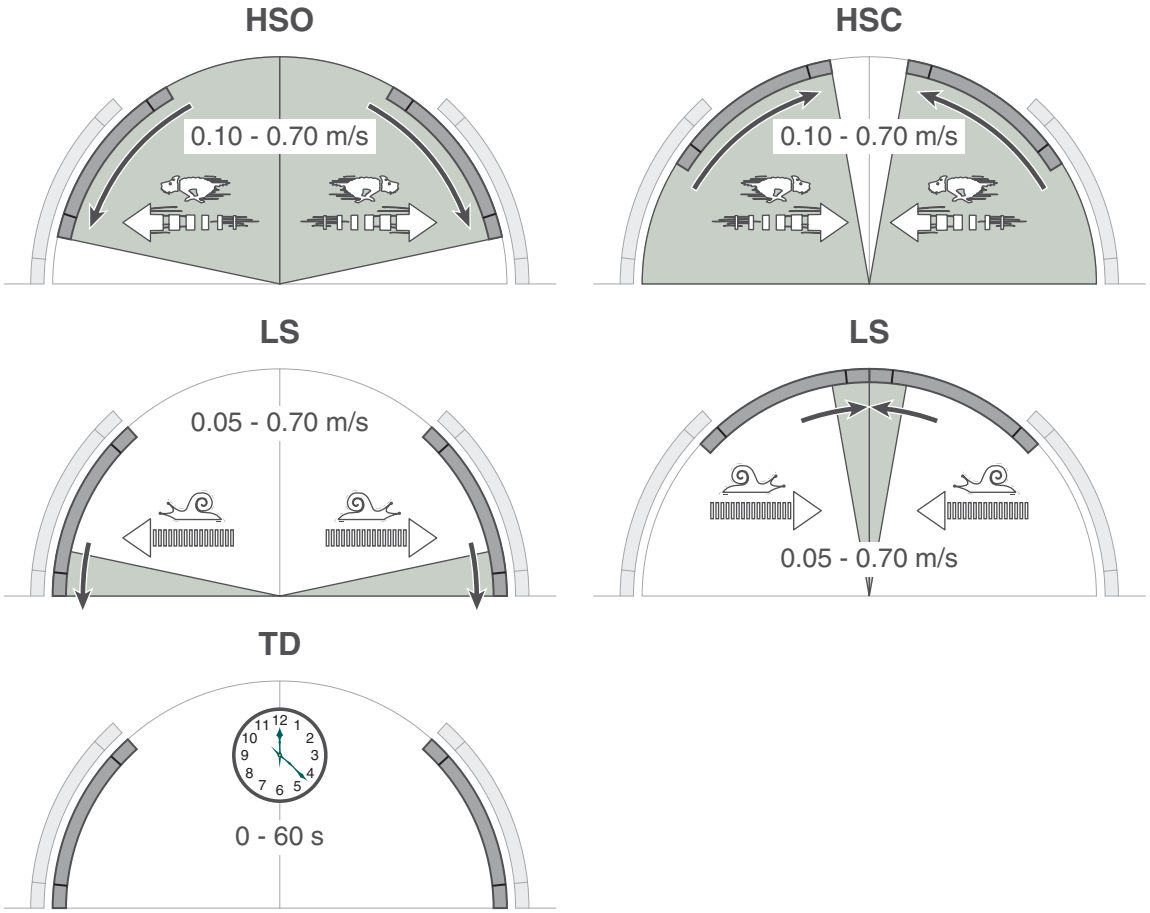
Function selector (FS)	OFF	ON
1. Motor direction of rotation [Always to be set to OFF]	CCW	
2. Lock type (locked with/without power)	With LDP ILL-00200	Without LD LDB ILL-00201
3. Lock release [To be set to ON if electric lock is installed]	No	Yes
4. Presence detection type (normally open/closed) [Applies in common for the terminals 11, 12 and 13 on the control unit CU] CU 14 13 Stop impulse 12 Presence impulse 2 11 Presence impulse 1 10 ILL-00202	NO ILL-00203	NC ILL-00204
5. Emergency unit type [If no emergency unit is installed the parameter should be set to OFF]	EEU ILL-00205	MEU ILL-00206
6. Emergency unit monitoring [To be set in accordance with local authority requirements]	No	Yes
7. Sensor monitoring [To be set in accordance with local authority requirements]	No	Yes
8. Hold force on closed door (0 N / 45 N) [Setting ON always recommended]	No	Yes

Note! Perform a reset after any FS adjustment to ensure proper configuration.



8.2.4 Potentiometers and learn button

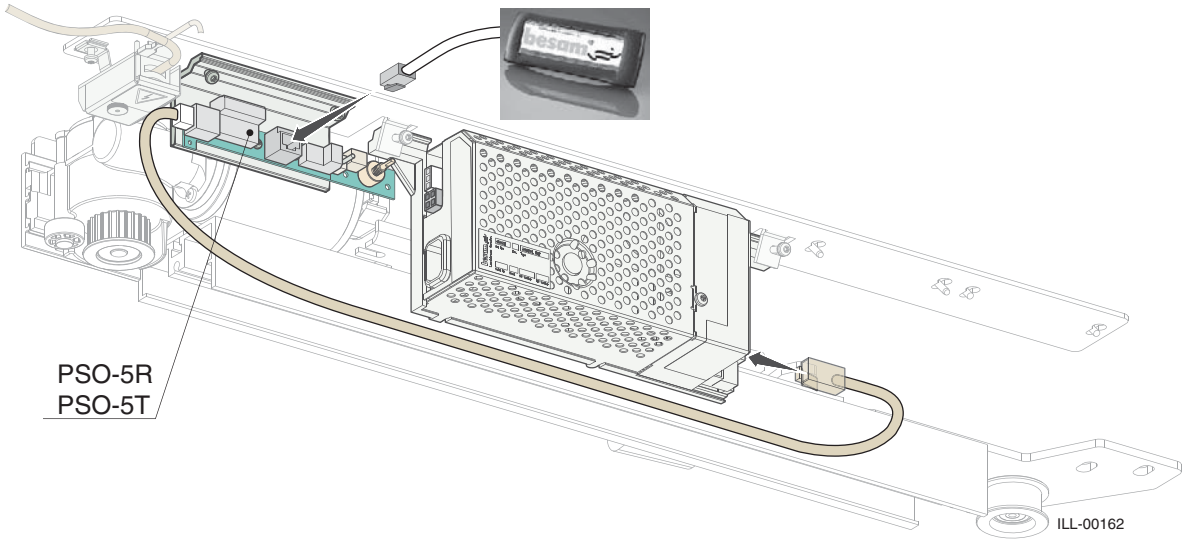


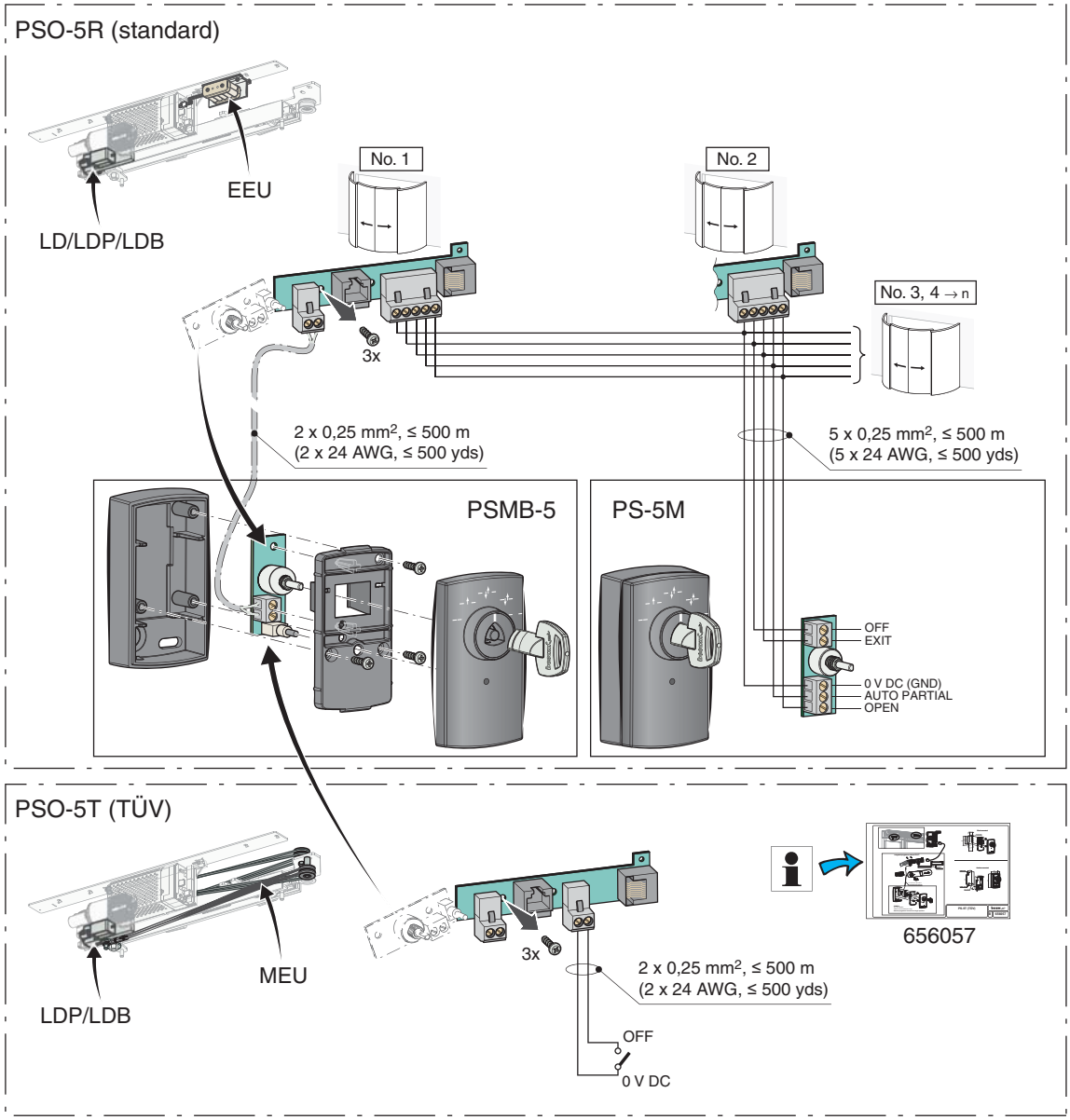


The potentiometers are factory set to approx. 50% of the adjustment range.

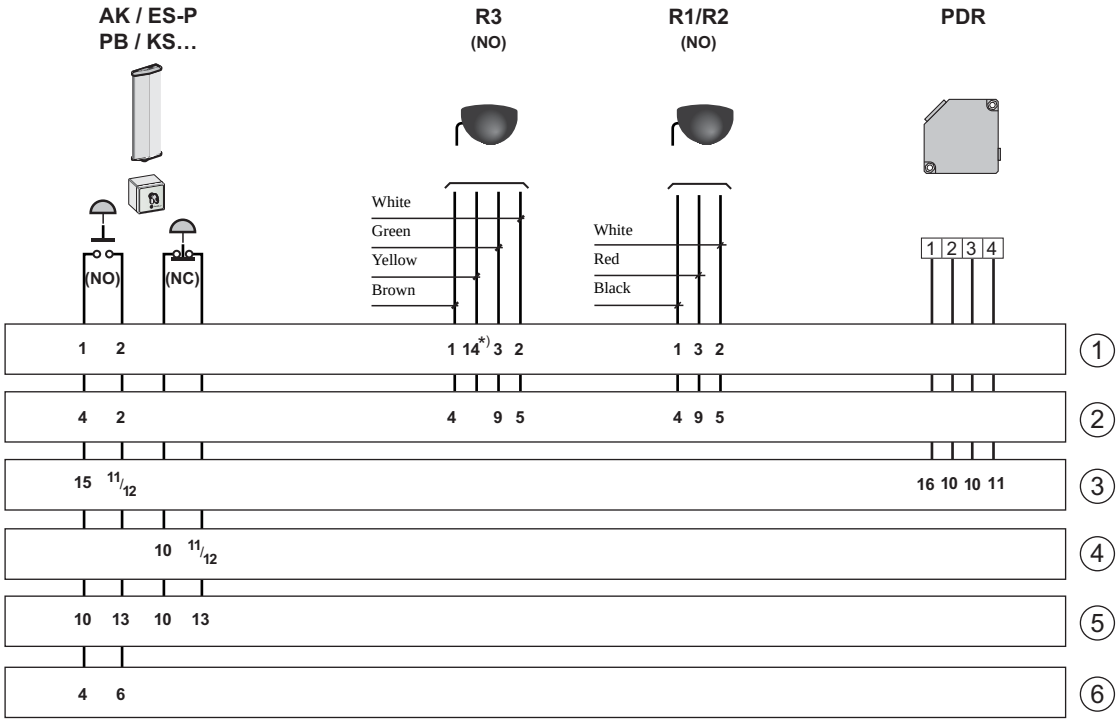
Note! The speed applies to a single leaf.

8.2.5 Connection of programme selectors





8.2.6 Connection of activation units



*) Only to be connected if inner impulse monitoring is required, FS-7 = ON

(1) Inner impulse
(2) Outer impulse
(3) Presence impulse, NO
(4) Presence impulse, NC
(5) Stop impulse
(6) Key impulse / (Emergency open impulse, EEU required)

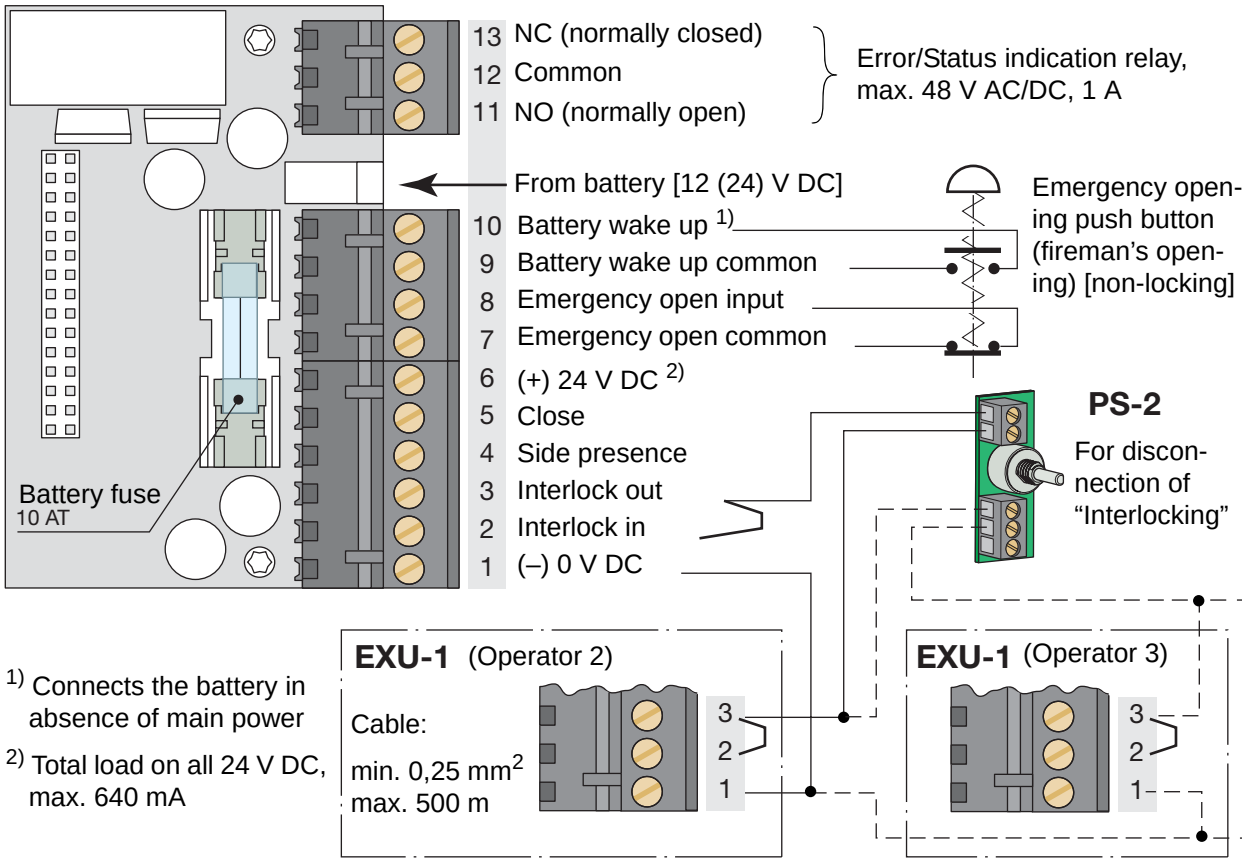
Note! If necessary, adjust the sensor detection fields to meet the specific application and regulation requirements.

8.3 Extension unit

When functions beyond those implemented on the main control unit are required, an extension unit is available, EXU-1. The unit is factory-installed on top of the control unit.

8.3.1 Extension unit, EXU-1

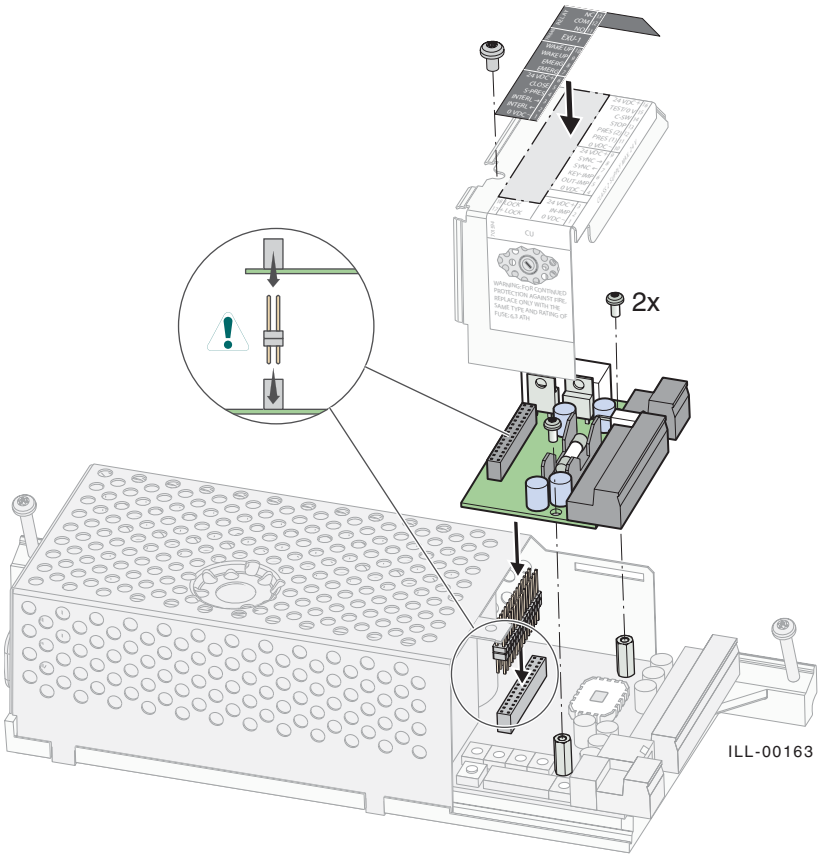
Following functions can be obtained with this unit:



1) Connects the battery in absence of main power

2) Total load on all 24 V DC, max. 640 mA

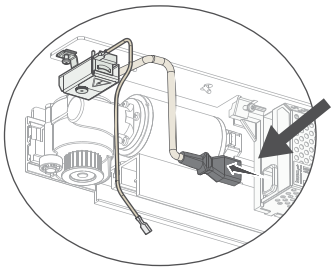
8.3.2 Replacement of the extension unit EXU-1



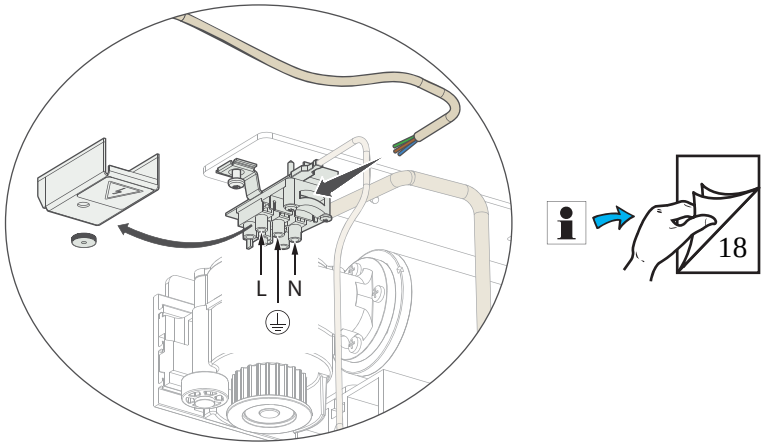
9 Start-up

After installing the automatic door, the Start-up and adjustment must be carried out in the following order (see also “Electrical connections” on page 18).

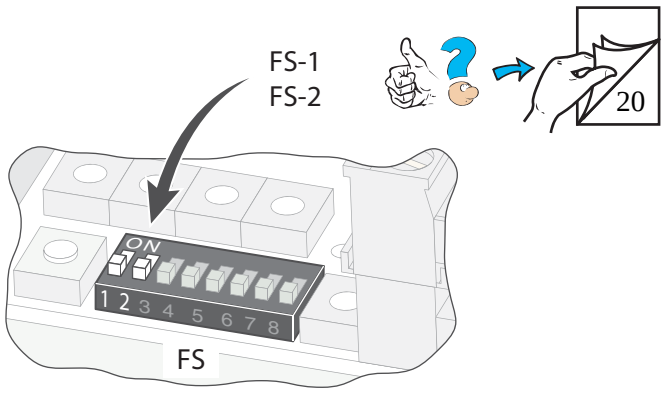
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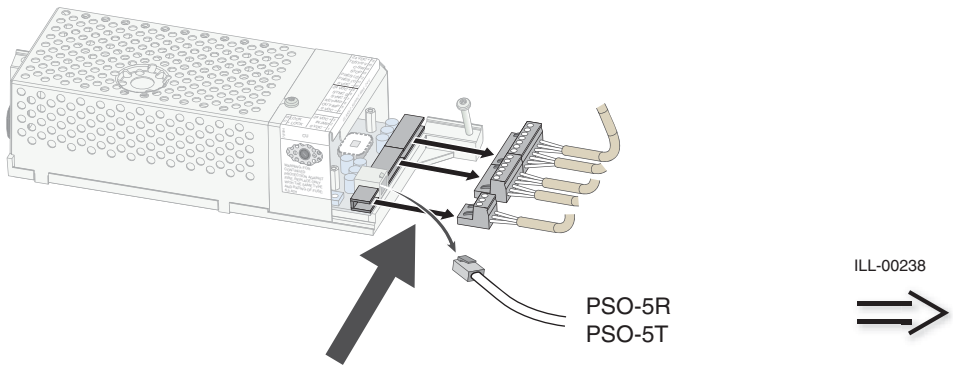
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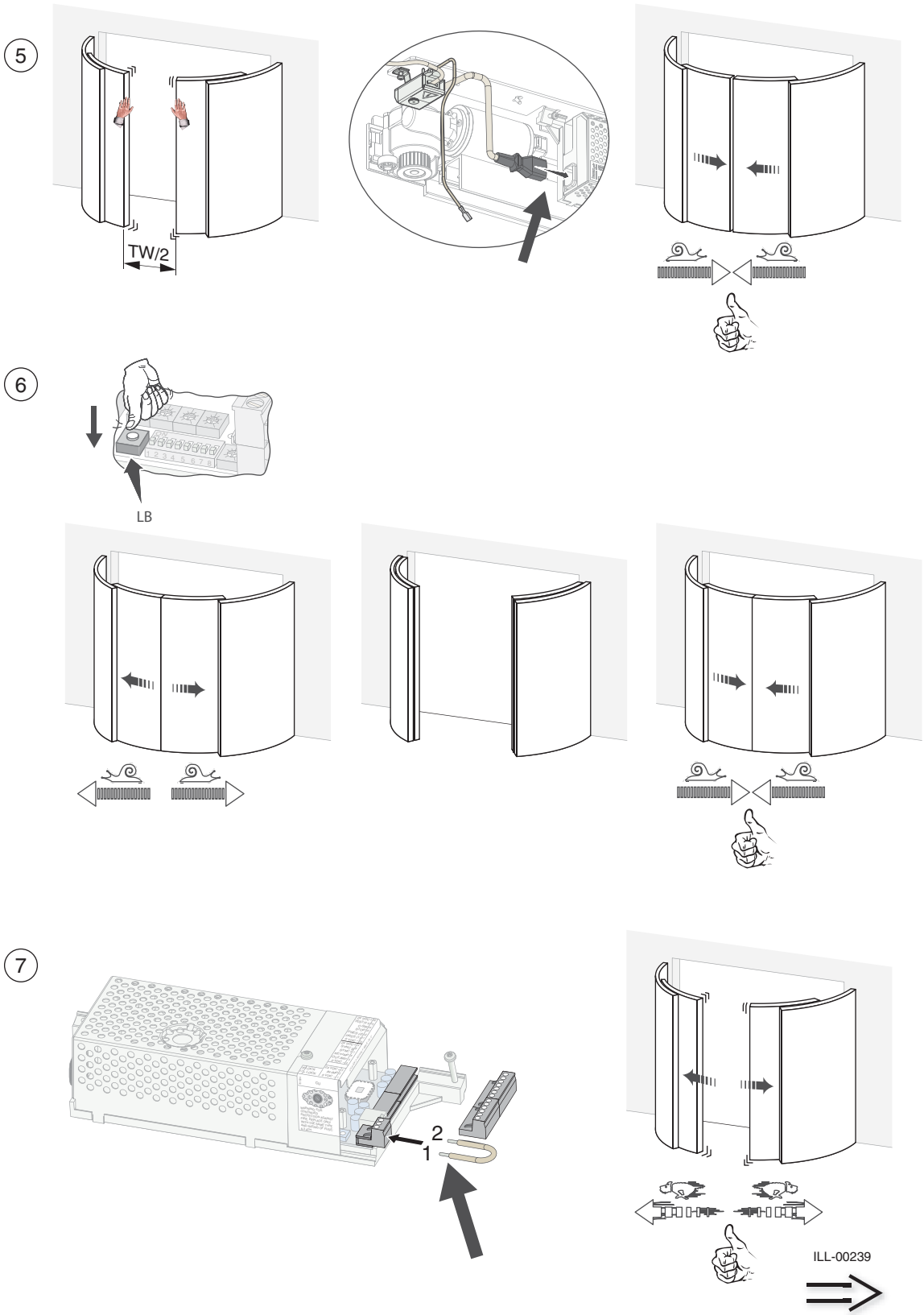


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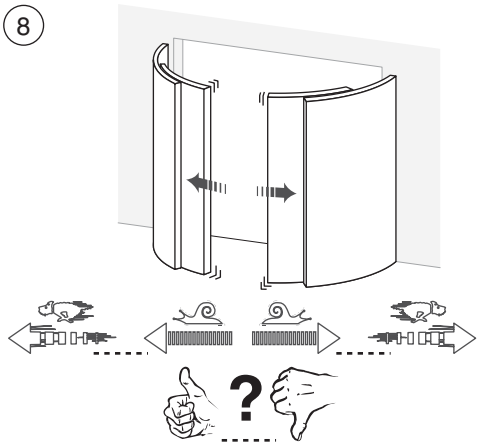


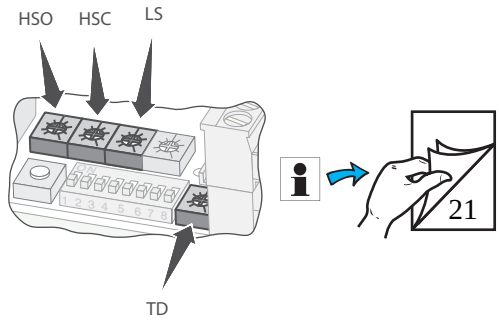


If necessary, adjust the door speeds to meet the specific application and regulation requirements.

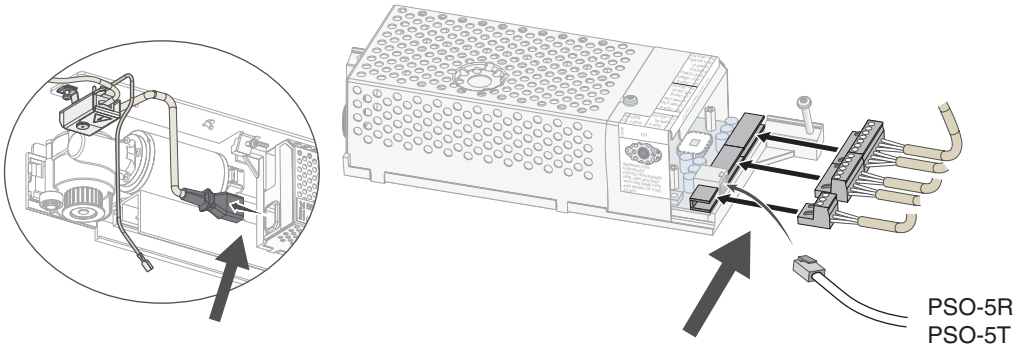
Note! The “low speed” is self adjusting to optimal operation if the “low speed parameter” is set to **max**. Depending on authority or installation requirements the “low speed” can be further reduced.

8

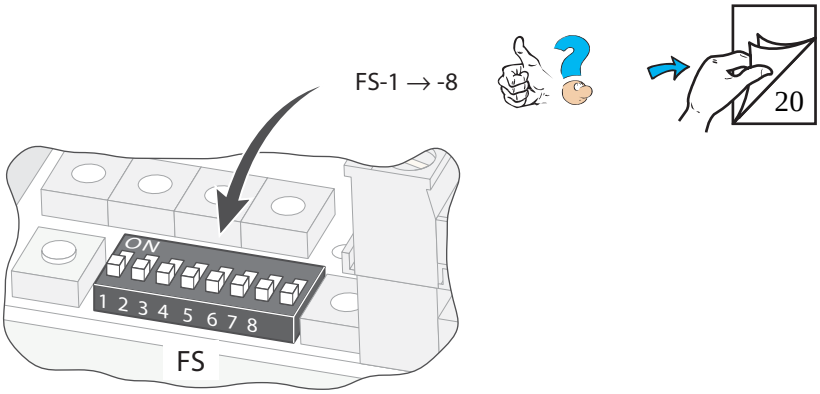




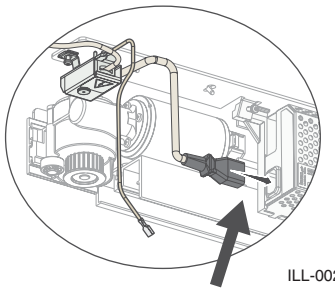
9



10



11



Note! Perform a reset after any FS adjustment to ensure proper configuration.

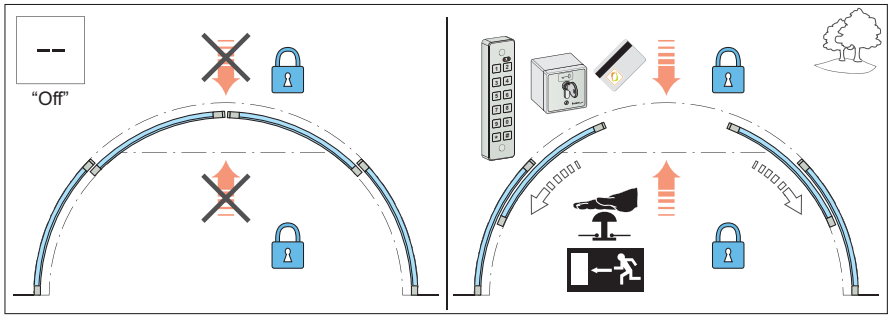
Note! Check that the installation complies with valid regulations and requirements from the authorities.

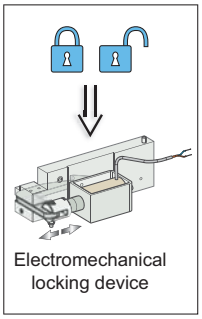
10 Programme selectors

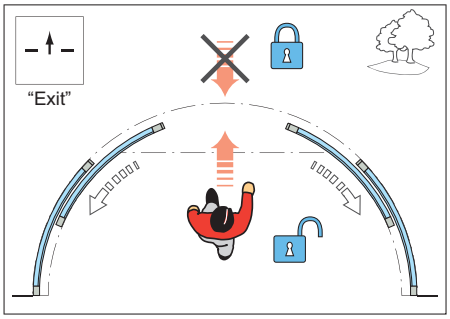
Note! The key must always be removed on emergency escape doors after changing settings.

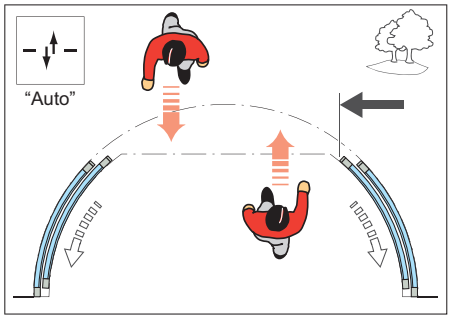
Note! In “Off”: The door can be manually pulled open if no mechanical/electromechanical lock is fitted

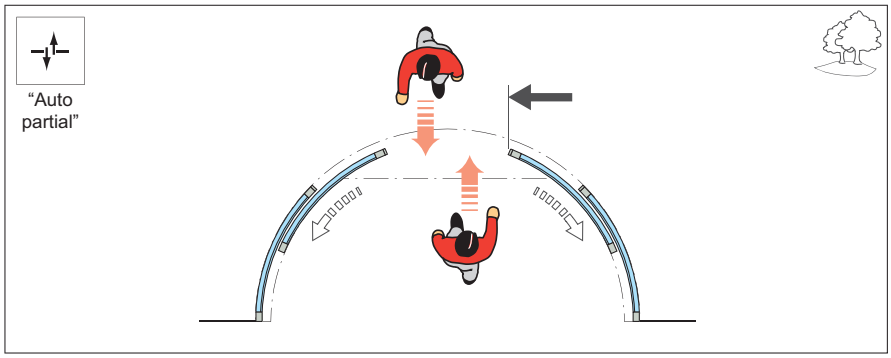


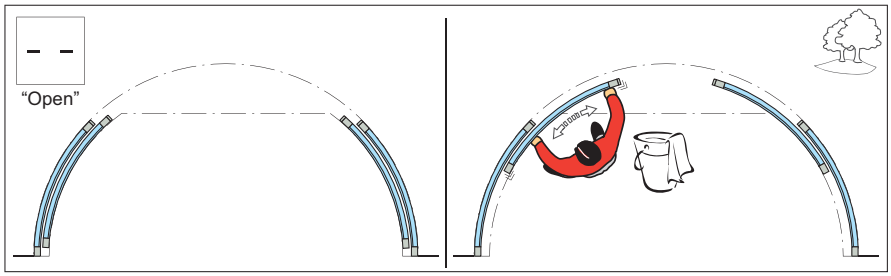


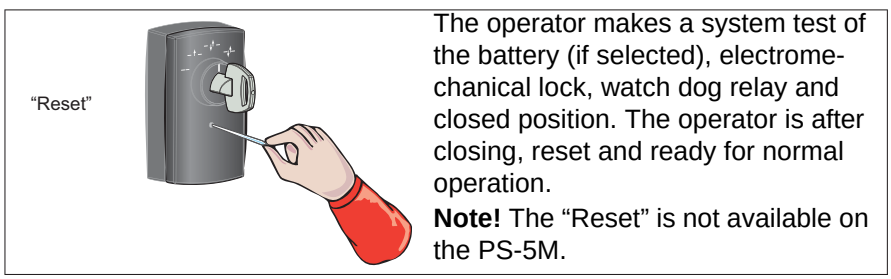












The operator makes a system test of the battery (if selected), electromechanical lock, watch dog relay and closed position. The operator is after closing, reset and ready for normal operation.

Note! The “Reset” is not available on the PS-5M.

11 Troubleshooting

Before starting the troubleshooting, check that the programme selector setting is correct and then reset the operator. Start the troubleshooting by checking the mechanical and electrical parts of the operator in the following order.
The control unit, emergency unit and electromechanical lock are fixed with brackets in the support beam. To replace, the complete unit is to be loosened and replaced.

11.1 Mechanical checking and remedies

Disconnect the mains power. Unlock all mechanical locks. Pull the door leaf manually and check that the door can be easily moved over the complete sliding track/floor guide. If the door leaf stops or is hard to move, the reason may be sand, stones, rubbish etc. in the floor guide. The door leaf may also be jamming on the floor or on the draught excluders. Clean the floor guide, adjust the door leaf height/depth or take other necessary measures e.g. replacement of wear parts until the door leaf is running smoothly when manually operated.

11.2 LED indication and CT Error codes

The control unit is equipped with a light emitting diode LED for error indication. By means of the configuration tool CT, a more detailed error description (CT error codes) can be obtained. See also separate manual, IAA004.

11.2.1 Normal operation/Non-critical errors

During normal operation and for non-critical errors the LED on the control unit is illuminated.

11.2.2 Power failure (no error code)

If the LED is extinguished check the mains power, power supply cable, CU-fuse 6,3 ATH and perform a reset. If the problem remains replace the control unit.

11.2.3 LED indication

- **Sensor error**
  1 fast flash (0,2s), pause (1,0s) etc.
- **Emergency Unit error**
  2 fast flashes (0,4s), pause (1,0s) etc.
- **CU error**
  3 fast flashes (0,6s), pause (1,0s) etc.
- **Motor/Encoder error** 
  4 fast flashes (0,8s), pause (1,0s) etc.
- **Lock error**
  5 fast flashes (1,0s), pause (1,0s) etc.
- **Motor Temperature High**
  1 slow flash (1,8s), pause (0,2s) etc.
- **Normal operation/Non-Critical errors**
  Illuminated

11.2.4 Sensor Error

LED indication: 1 fast flash (0,2s), pause (1,0s), etc.

CT error code	Reason	Remedy
Presence Impulse Error	• The control unit doesn't get a test answer, within 200 ms, from the activation unit.	• Make sure that the monitoring output is connected and the connections are OK.
		• Replace the presence activation unit.
Inner Impulse Error	• The control unit doesn't get a test answer, within 200 ms, from the activation unit.	• Make sure that the monitoring output is connected and the connections are OK.
		• Adjust sensor field so that the sensor can give a test answer.

11.2.5 Emergency Unit Error

The door is opened and stays open

LED indication: 2 fast flashes (0,4s), pause (1,0s), etc.

CT error code	Reason	Remedy
Emergency Action Timeout	• The door is prevented from performing its battery test within 30 seconds, due to low opening speed or high friction.	• Make sure that the High Speed Opening is above 0.2 m/s, and that the door is moving with low friction.
Emergency Unit Error	• The door is prevented from performing its rubber band test within 10 seconds, due to broken or weak rubber band or high friction.	• Check rubber band tension and make sure that the door can open to fully open position.
	• The battery voltage drops below 9 V, due to low capacity.	• Charge or replace battery.
Battery Error	• The battery is disconnected or short-circuited.	• Make sure that the cables are OK and connected.
		• Check the battery fuse.
		• Charge or replace battery.

11.2.6 CU error

LED indication: 3 fast flashes (0,6s), pause (1,0s), etc. ●●●▭▭▭●●●▭▭▭●●●

CT error code	Reason	Remedy
RAM Error	• Internal RAM memory error.	• Reset, and if the problem remains, replace the control unit.
ROM Error	• Internal ROM memory error.	• Reset, and if the problem remains, replace the control unit.
EEPROM Error	• Serious internal EEPROM memory error.	• Reset. • Download a saved parameter set or DEFAULT parameter set and perform a reset. If the problem remains, replace the control unit.
EEPROM Critical Write Error	• Internal write EEPROM memory error. This error mainly occurs when it's impossible to change a configuration parameter.	• Reset. • Try to change the configuration parameter that caused the problem, and if the problem remains, replace the control unit.
SMPS Over Voltage	• The internal link voltage has for some reason increased to above 47 V.	• Reset, and if the problem remains, replace the control unit.
A/D Converter Error	• The internal A/D Converter or multiplexer is broken.	• Reset, and if the problem remains, replace the control unit.
Watchdog Relay Error	• It's not possible to disconnect the motor bridge and lock with the watchdog relay.	• Reset, and if the problem remains, replace the control unit.
Hardware Watchdog Error	• It's not possible to disable the motor bridge.	• Reset, and if the problem remains, replace the control unit.

11.2.7 Motor / Encoder error

The motor and lock power are disconnected.
LED indication: 4 fast flashes (0,8s), pause (1,0s), etc.

CT error code	Reason	Remedy
Encoder Error	• The encoder, encoder cable, or Motor cable is damaged.	• Make sure that the encoder cable and the motor cable is connected.
	• Wrong Motor type is selected.	• Check Motor Type configuration with the CT-Tool.
Motor Current Error	• The Motor cable or Encoder cable is damaged.	• Make sure that the encoder cable and the motor cable is connected.
	• Wrong Motor type is selected.	• Check Motor Type configuration with the CT-Tool.

11.2.8 Lock error

The motor and lock power are disconnected.
LED indication: 5 fast flashes (1,0s), pause (1,0s), etc.

CT error code	Reason	Remedy
Lock Failure	• The lock or something else was preventing the door from opening the first 14 mm from closed position.	• Make sure that the lock is operating without friction.
		• Make sure that Hold Force and Lock Release are set correctly.

11.2.9 Motor Temperature High

The door is opened and stays open.
LED indication: 1 slow flash (1,8s), pause (0,2s), etc.

CT error code	Reason	Remedy
Motor Temperature High	• The duty cycle of the door is to high for the current speed settings and hold open time.	• If the motor is warm, put the door in PS OPEN and wait for at least 1 minute. Reduce speeds and increase hold open time.
	• The heavy-duty motor is replaced with a normal duty motor.	• Put the door in PS OPEN and wait for at least 5 minutes. **Note!**This error is not removable by reset, only by Setting the door in PS OPEN with the power on.

11.2.10 on-Critical errors

These errors don't influence the door operation but are logged in the error log, and can only be displayed by means of the CT-Tool.
LED indication: Illuminated

CT error code	Reason	Remedy
Communication Error	• The cable to the CT-Tool was removed before performing “Disconnect” from the CT-Tool.	• Connect the CT-Tool and “Disconnect” from the CT-Tool before removing the cable.
EEPROM Access Error	• The EEPROM queue is full.	• To many events to log in the event log. Reduce the number of events to log in the event log configuration.
EEPROM Non-critical Write Error	• The control unit cannot write error log, event log, or service log information to the EEPROM memory.	• Reset, and if the problem remains, replace the control unit if it is important to read log information.
Program Flow Error	• Internal program error.	• Replace the control unit.

11.3 After remedy or replacement the operator has to be checked as follows:

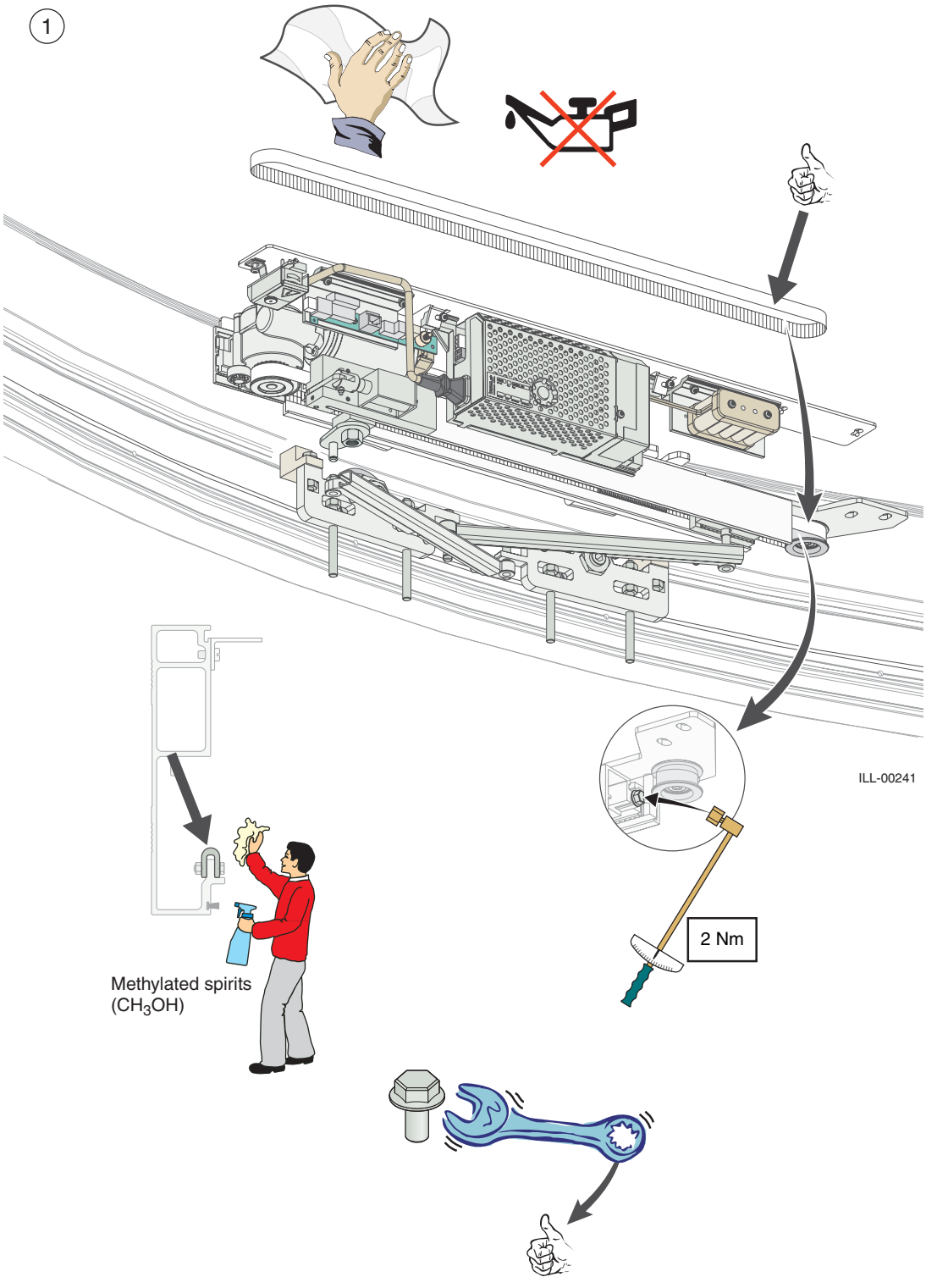
1. Study the door movement and adjust the functions to the values required for a smooth door operation.
2. Check that correct functions and values have been selected for the installed accessories and that the installation complies with valid regulations and requirements from the authorities.
3. Clean the cover and the doors.

12 Maintenance/Service

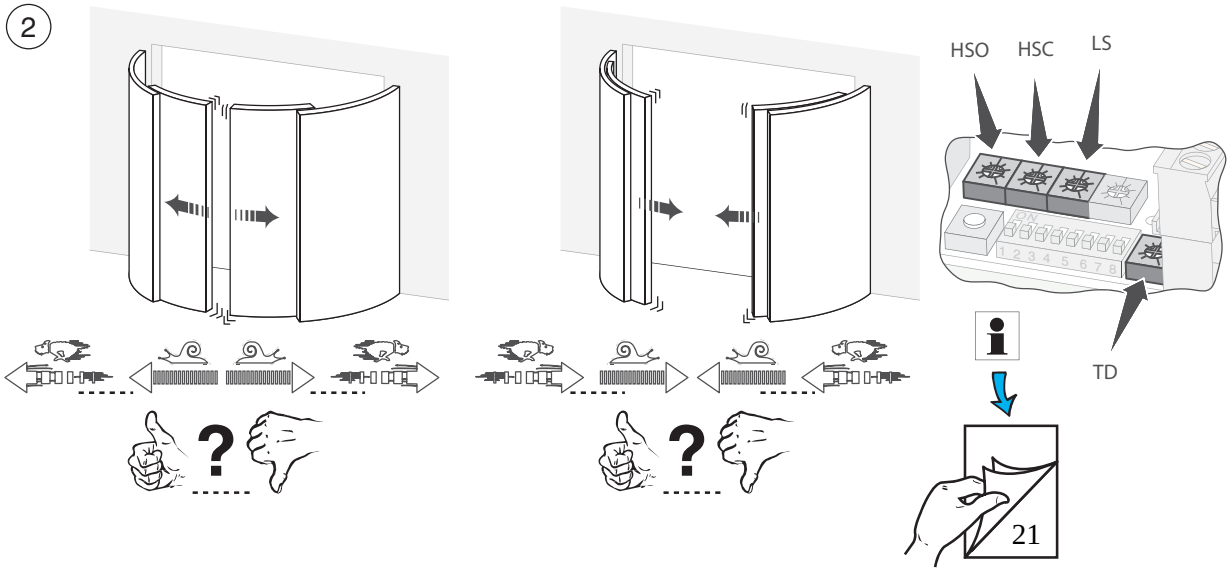
Automatic door installations must be subjected to regular maintenance, the frequency of which is governed by the environmental conditions and density of traffic.

According to regulations inspection of emergency escape doors is required at least twice a year, carried out by an authorized service centre.

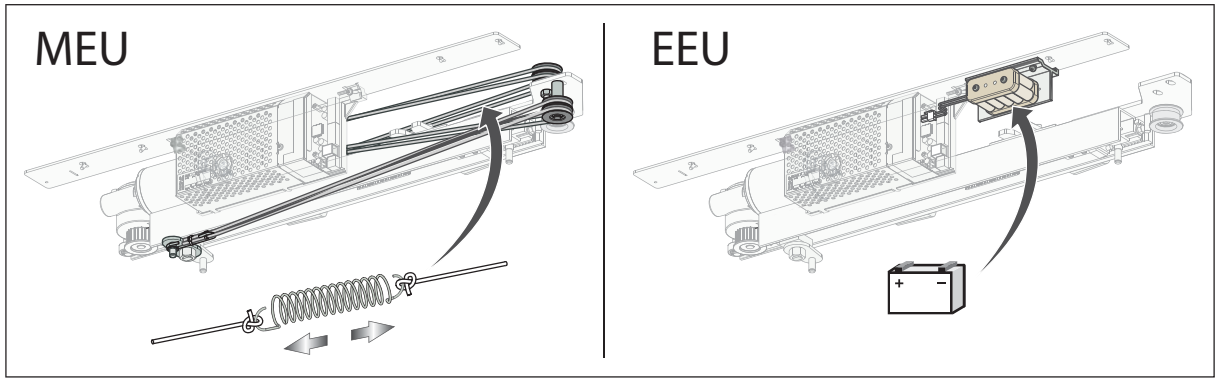
Note! The force for redundant opening with MEU is created with help of elastic bands. The material for these bands may be slowly degraded if exposed to lubricants.



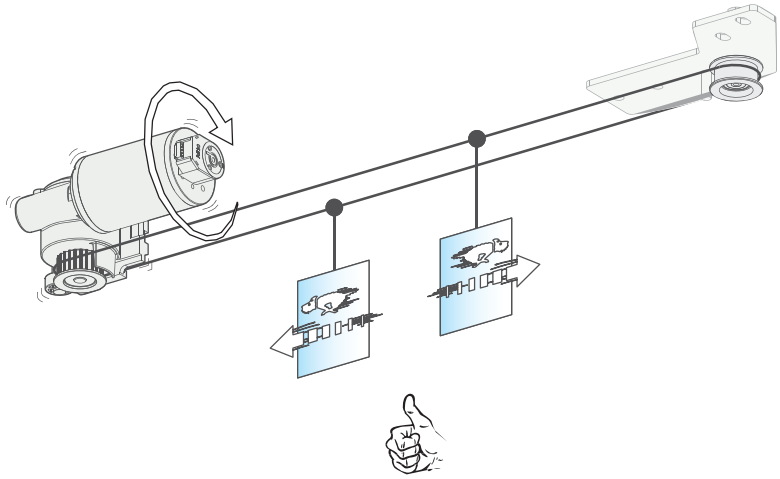
2

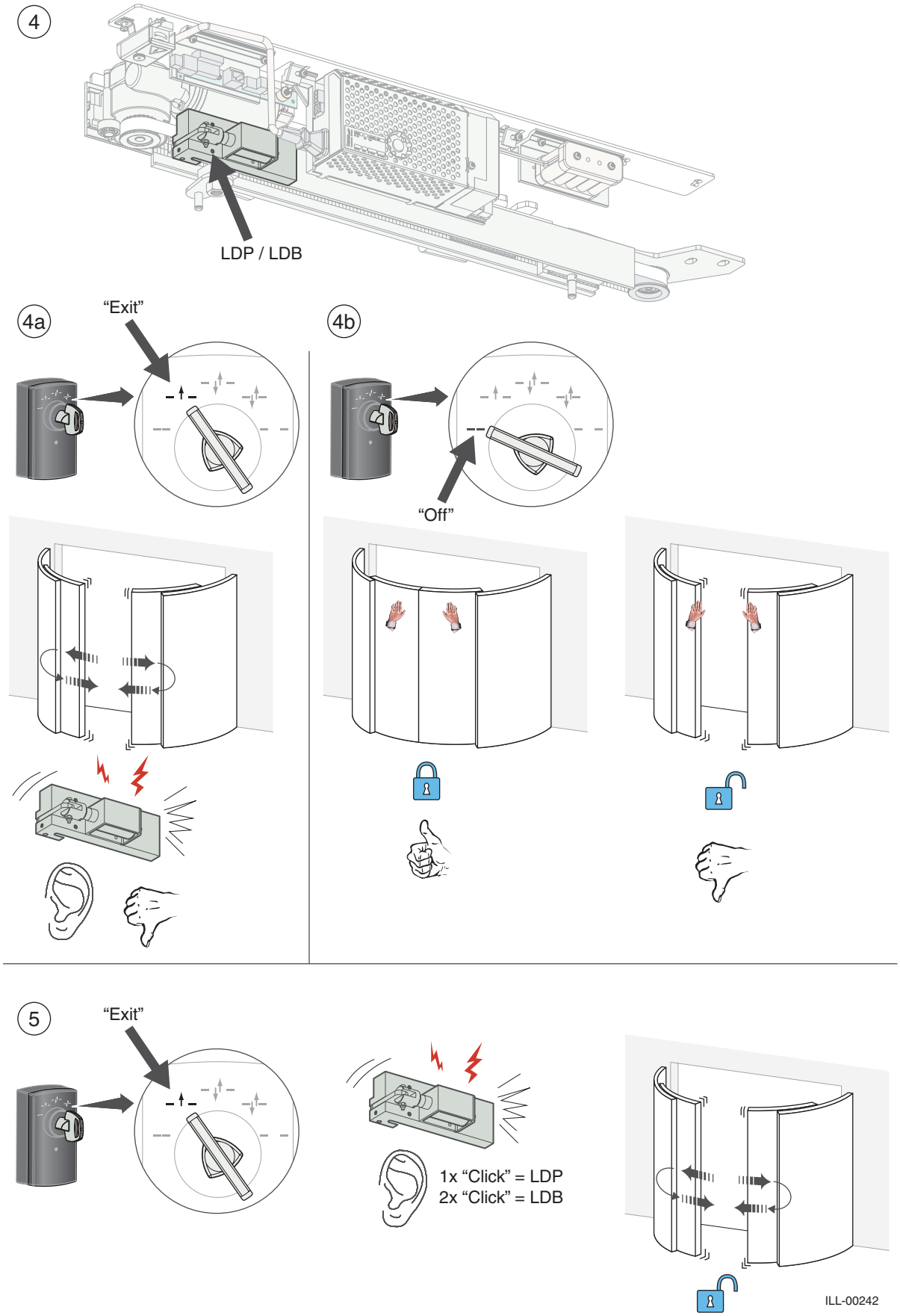


3



ILL-00243




4
LDP / LDB
4a
"Exit"
4b
"Off"
5
"Exit"
1x "Click" = LDP
2x "Click" = LDB
ILL-00242

13 Accessories

Besam CMD, can be equipped with the following standard or optional equipment. If you wish to upgrade your operator, or add some extra equipment, contact your Besam representative.

Cover	Made in clear anodized aluminium as standard. Paint finished in RAL colours or anodizing optional.
Activation units	See separate manuals or installation drawings.
Programme selector PSMB-5 and **PS-5M**	See page 29 and separate installation drawing 656 037.
Programme selector PS-2	See separate installation drawing 656 058.
Electromechanical locking device LD (locked without power), **LDP** (locked with power) and **LDB** (bi-stable)	See separate installation drawing 1002012.
Micro switch kit LSK	For indication of door and lock position.
Electrical emergency unit EEU	See page 11.
Emergency closing with repeated closing	If the door is opened by hand after an electrical emergency closing, it will close again (LSK required), see separate installation drawing 656 006.
Mechanical emergency unit MEU	See page 11 and separate drawing 1001810.
Interlocking	Used between two operators when the first operator must close before the other one can open (typical to reduce energy losses and not for security reasons).
Convenience battery UPS	Stand-by supply, which gives continued operation during short power failure, see separate installation drawing 656 056.
Emergency opening	(Fireman's opening) opens the door in any programme selector setting.
External error indication	Obtained if a lamp or a buzzer is connected.
Key switches KS-F (flush) /**KS-S** (surface)	Used to give opening impulse to the door in any programme selector setting. With EEU also during power failure, see separate installation drawing 655 967.
Push button PB	Used to give opening impulse to the door, see separate installation drawing 656 005.

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